

Electrical Circuit Systems are Hamiltonian

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An ordinary non-dissipative circuit, cooled to cryogenic temperatures, becomes a finite-dimensional quantum mechanical system. Identifying its classical Hamiltonian is a prerequisite for quantization, yet standard approaches treat the canonically conjugate magnetic flux and electric charge variables asymmetrically, or fail outright when neither admits a global Lagrangian description. By combining topological graph theory with symplectic geometry, we translate a circuit into a mathematical object whose null structure is precisely its gauge redundancy, and whose symmetry under flux-charge exchange is manifest from the start. The degeneracies that obstruct classical formulations become the central objects of study; the circuit's semantics are baked into a graph-theoretic translation. Following Osborne and Lucas, we develop the formalism from first principles, recover the linear LC theory as a special case, and prove that every finite non-dissipative circuit with smooth constitutive relations is a Hamiltonian dynamical system.

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1. INTRODUCTION

A non-dissipative electrical circuit is a physical object. It consists of wires, capacitors, inductors, and (at extreme low temperatures) Josephson junctions and quantum phase slip elements arranged on a board. Quantizing a circuit means identifying its Hilbert space and Hamiltonian, which are prerequisites in many applications. Standard quantization procedures often rely on identifying the Hamiltonian for a dynamical system, a task that is straightforward in some circuits, and subtle in others. This paper develops a mathematical framework that renders the construction natural.

Throughout the discussion, the guiding philosophy is that a circuit should not be treated as a piece of hardware upon which physical constraints are imposed. Instead, we inherit from the work of Osborne and Lucas [6, 7], the approach of treating circuits a priori as mathematical objects, wherein through a constructive graph-theoretic approach, every physical constraint becomes a topological feature. Topological and algebraic graph theory, together with symplectic geometry, determine the Hamiltonian entirely from the underlying structure.

What allows this to happen is the careful analysis of the gauge freedoms within a circuit, both semantically, and as graphical structures. Assumptions of linearity will provide us with an insight into the general procedure: identifying null directions to project into a phase space that attains canonical coordinates. When we push beyond linearity to superconducting circuits, periodic relations prevent a flux-only Lagrangian description. A more nuanced mechanism, that of holding flux and charge on equal footing from the outset, will provide a symmetric treatment that reveals an equivalence between gauge freedoms and homogeneous graph structures. Another graphical object—the spanning tree—will then provide a constructive derivation of the canonical conjugate pairs.

The goal is to show that every well-posed circuit admits a Hamiltonian dynamical description. We assume the circuit is finite, connected, non-dissipative, planar, with smooth constitutive relations.

Prior Work and Notation/Conventions

This paper follows closely the development of Osborne [6] and Osborne-Lucas [7]. The first 2023 paper introduced the symplectic-geometric approach to circuit quantization revolving around the reduced capacitive incidence matrix $\Omega = A_C^\top$; the 2024 paper identified the asymmetric treatment inherent in this choice and replaced Ω with the flux-charge symmetric connection matrix M .

Our presentation synthesizes both: we begin with the Ω -based formulation implicit in the linear LC proof, and then develop the M -based improvement. The Ω -based formulation of [6] is perfectly capable of quantizing all circuits we consider, including the Dualmon—it is not broken, but biased. The consequences of this, as well as its symmetric fix, will be explained in [Section 6.1](#).

This discussion weaves one cohesive narrative leading to the final result. Along the way, we have taken the opportunity to provide additional arguments and fill in mathematical details—such as proofs of exactness, a symmetric model explanation of spanning trees, invariance under spanning tree choice, derivations of the Poisson bracket, representative examples, and a reverse reading.

We will use bra-ket notation for vectors in graph-associated vector spaces. The bra-ket syntax depicts the mapping directions of matrices such as A , M , and Ω , and makes duality explicit: transposes are written as $A^\top$, where $*$ is reserved for the dual. Note that our choice of orientation for the incidence matrix is $A \in \mathbb{R}^{|V| \times |E|}$. This differs in transpose from the convention $A_{e,v}$ used in [6, 7].

Because both fluxes and charges live on multiple natural indices (vertices, edges, faces, loops), we denote each ket with a subscript: $|\phi_V\rangle \in \mathbb{R}^{|V|}$ for node fluxes and $|\phi_E\rangle \in \mathbb{R}^{|E|}$ for branch fluxes;

analogously $|q_E\rangle \in \mathbb{R}^{|E|}$ for branch charges and $|q_F\rangle \in \mathbb{R}^{|F|}$ for face charges, which on a planar embedding coincide with the loop-charge coordinates that will appear in the symplectic pairing. For canonical coordinates $(\Phi_t, Q_t)_{t \in T}$ identified in [Section 8](#), the subscript t runs over branches in a tree T . To avoid confusion, time will be denoted with τ .

Finally, we will use the Pauli matrices standard in quantum mechanics: $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

2. DYNAMICAL SYSTEM FORMALISM

The purpose of this section is to establish a language through which different characterizations of “dynamics” can be compared. The formalism largely follows Goldstein [\[2\]](#) and McDuff-Salamon [\[5\]](#).

Each definition is formalized in isolation, and it is the subtle interplay between them that we are concerned with. Circuits naturally define a dynamical system through their elements’ constitutive relations and Kirchhoff’s laws, but such a system of equations is not necessarily a priori variational or symplectic. Conversely, Lagrangian and Hamiltonian systems are defined through different constraints—an action functional, or a nondegenerate bracket—which are not automatically satisfied by an arbitrary system of equations. Through careful definitions, we will come to understand at a structural and graph-theoretic level how and when these dynamical systems coincide.

Definition 2.1 (Dynamical System). A dynamical system is a system of differential equations

$$\dot{x} = F(x), \quad x \in \mathbb{R}^N, \quad (2.1)$$

for a smooth vector field $F : \mathbb{R}^N \rightarrow \mathbb{R}^N$. A trajectory is a curve $x(\tau)$ satisfying the system for all τ .

Definition 2.2 (Circuit Dynamical System). A circuit dynamical system is a dynamical system whose equations of motion are obtained from a well-posed circuit by applying Kirchhoff’s laws and the constitutive relations of the elements. [Section 3](#) will formalize this definition in a constructive graph-theoretic manner.

Definition 2.3 (Linear Dynamical System). A dynamical system $\dot{x} = F(x)$ on $\mathbb{R}^n$ is linear if F is a linear map: there exists a matrix $M \in \mathbb{R}^{n \times n}$ such that $F(x) = Mx$ for all x . Differentiating yields

$$\ddot{x} = M^2 x. \quad (2.2)$$

If M has a block structure, this is reflected in M^2 . For example, if $M = \sigma_x \otimes X$, then $M^2 = I \otimes X^2$ is block diagonal. Consequently, the second-order equation decouples into independent equations

$$\ddot{x} = X^2 x \quad (2.3)$$

on each block. This decoupling yields independent second-order circuit dynamics.

The definitions thus far are structural: they describe the shape of an equation of motion without committing to where it comes from. To talk about Lagrangian systems, we need a richer object—a functional on curves whose stationary points are the equations of motion. This creates the geometric scaffolding (action, symplectic form, Poisson bracket) that quantization will eventually require.

Definition 2.4 (Action Functional). Let $\mathcal{C}$ be a space of sufficiently smooth curves $x : [\tau_0, \tau_1] \rightarrow \mathbb{R}^n$. An *action functional* is a map $S : \mathcal{C} \rightarrow \mathbb{R}$ of the form

$$S[x] = \int_{\tau_0}^{\tau_1} L(x(\tau), \dot{x}(\tau), \tau) d\tau, \quad (2.4)$$

where $L : \mathbb{R}^n \times \mathbb{R}^n \times [\tau_0, \tau_1] \rightarrow \mathbb{R}$ is a smooth function called the *Lagrangian*.

To say that a curve makes S stationary, we need a derivative of S —but S is a function of an entire curve $x_i(\cdot)$, not of a finite list of variables, so the ordinary partial derivative is not the right object. The replacement is the *functional derivative*.

Definition 2.5 (Functional Derivative). Let $S : \mathcal{C} \rightarrow \mathbb{R}$ be an action functional. The functional derivative of S with respect to the curve x_i , written $\frac{\delta S}{\delta x_i}$, is the function of τ characterized by

$$\int_{\tau_0}^{\tau_1} \frac{\delta S}{\delta x_i}(\tau) \eta(\tau) d\tau = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} S[x_1, \dots, x_i + \varepsilon \eta, \dots, x_n] \quad (2.5)$$

for every smooth perturbation $\eta : [\tau_0, \tau_1] \rightarrow \mathbb{R}$ vanishing at the endpoints.

Essentially, $\delta S / \delta x_i$ measures how S responds to a local wiggle of the curve x_i at time τ , analogous to how $\partial f / \partial x_i$ measures how f responds to a wiggle of the coordinate x_i . The difference is that the input is now a curve and the output is a function of τ .

Example 2.6 (The Euler–Lagrange operator). For an action $S[x] = \int_{\tau_0}^{\tau_1} L(x, \dot{x}, \tau) d\tau$, integrating the variation $\left. \frac{d}{d\varepsilon} \right|_0 S[x + \varepsilon \eta]$ by parts once and using boundary conditions $\eta(\tau_0) = \eta(\tau_1) = 0$ yields

$$\frac{\delta S}{\delta x_i} = \frac{\partial L}{\partial x_i} - \frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}_i}. \quad (2.6)$$

The condition $\delta S / \delta x_i = 0$ is therefore the familiar *Euler–Lagrange equation*

$$\frac{d}{d\tau} \frac{\partial L}{\partial \dot{x}_i} - \frac{\partial L}{\partial x_i} = 0. \quad (2.7)$$

Definition 2.7 (Lagrangian Dynamical System). Let S be an action functional depending on n variables $x_1, \dots, x_n$. Consider the system of differential equations

$$\frac{\delta S}{\delta x_i} = 0, \quad i = 1, 2, \dots, n, \quad (2.8)$$

where $\delta / \delta x_i$ is the functional derivative of [definition 2.5](#). If a system of differential equations satisfies (2.8) for some suitable choice of S , we say that the system is a *Lagrangian dynamical system*.

Remark 2.8 (No auxiliary variables). At a glance, it may appear that any dynamical system is Lagrangian, since one can always introduce auxiliary variables λ_i and define an action

$$S = \int \lambda^i (\dot{x}_i - F_i(x)) d\tau, \quad (2.9)$$

whose Euler–Lagrange equations (2.7) reproduce the original system $\dot{x}_i = F_i(x)$. However, the condition that a system be Lagrangian is non-trivial precisely because we insist that S depends only on the physical variables x_i and their derivatives—no artificial auxiliary or Lagrange-multiplier variables may be introduced, making (2.9) an invalid construction. Lagrange multipliers may well arise naturally in a physical system, but ad-hoc introductions to promote every equation of motion to a multiplier equation are not permitted. As we will see, the physical variables of a circuit are the fluxes and charges within the system, which are directly measurable time-integral quantities.

A Lagrangian description is one half of the story; the other is the Hamiltonian description, where the dynamics are encoded not by an action but by some energy function and a bracket operator on phase space. The two descriptions are often related by the Legendre transform when well-defined, but the Hamiltonian formulation is logically independent and turns out to be the right framework for quantization. We produce here a very abstract definition, one that we will return to often.

Definition 2.9 (Hamiltonian Dynamical System). A *bracket* on $C^\infty(\mathbb{R}^{2n})$ is any operation

$$\{\cdot, \cdot\} : C^\infty(\mathbb{R}^{2n}) \times C^\infty(\mathbb{R}^{2n}) \rightarrow C^\infty(\mathbb{R}^{2n}) \quad (2.10)$$

satisfying:

1. $\{f, g\} = -\{g, f\}$ (antisymmetry)
2. $\{f, g\} = 0$ for all f if and only if $g = 0$ (non-degeneracy)
3. $\{\cdot, \cdot\}$ is linear in both arguments (bilinearity)

Given a function $H \in C^\infty(\mathbb{R}^{2n})$, called the Hamiltonian, and a valid bracket $\{\cdot, \cdot\}$, a system of differential equations is a Hamiltonian dynamical system if

$$\dot{x} = \{x, H\}. \quad (2.11)$$

We work in a finite-dimensional linear setting, which allows us to omit the usual Jacobi and Leibniz properties; both hold automatically. “Linearity” here refers to the underlying algebraic space and is not to be confused with linear circuit constitutive relations.

The definition above tells us what a bracket *is*, but it does not tell us how to construct one. The next remark closes that gap: in finite dimensions a bracket is completely specified by a single matrix.

Remark 2.10 (Coordinate form of the bracket). Any bracket on $\mathbb{R}^{2n}$ in the sense of [definition 2.9](#) is induced by a nondegenerate antisymmetric matrix $\Omega \in \mathbb{R}^{2n \times 2n}$ via

$$\{f, g\} = (\nabla f)^\top \Omega \nabla g. \quad (2.12)$$

Choosing a bracket is therefore the same as choosing Ω , referred to as the symplectic matrix. A standard result from linear algebra (the canonical form for nondegenerate antisymmetric forms [\[1\]](#)) tells us that any such Ω can be brought to the block form

$$\Omega \sim J \otimes S, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (2.13)$$

for some symmetric invertible matrix S . Here $\otimes$ is the Kronecker product and $J = i\sigma_y$. The decomposition [\(2.13\)](#) is meant to demonstrate that the $2n$ -dimensional phase space naturally splits into n canonical pairs (q_i, p_i) .

With these definitions in place, the question driving the rest of the paper sharpens: given the dynamical system that a circuit produces, under what conditions is it Lagrangian [definition 2.7](#), and when is it also Hamiltonian [definition 2.9](#)?

3. CIRCUITS AS DIRECTED GRAPHS

3.1. Graph Setup

To study a circuit, we translate the physical object into one that is purely mathematical. Throughout this discussion, it is important that we not only understand the rigor behind the mathematical transformations we will make, but the motivation and physical interpretation of each step.

The translation proceeds through several manipulations. We first restrict to planar circuits, embed the resulting graph on a sphere (to make its graphical dual well-defined), assign the physical variables that will live on the faces and vertices, and then mechanically apply the constitutive relations and Kirchhoff’s laws edge by edge to obtain a full representative system of equations.

3.1.1. Step 1: Restriction to Planar Graphs

We restrict to circuits whose graph admits a *planar* embedding, i.e. a drawing in the plane with no crossing edges. Physically, this corresponds to circuits that can be drawn on a flat board without wire crossings. More importantly, planarity ensures the existence of a dual graph, so that loop and node variables can be treated equally. Eventually, this will allow us to identify natural canonical variables and connect circuit dynamics to Hamiltonian structure.

Definition 3.1 (Circuit Graph). A circuit graph is a tuple $G = (V, E, \sigma)$ where

- V is a finite non-empty set of nodes (vertices);
- E is a finite non-empty set of directed edges, known as branches, between nodes:

$$e = (u, v) \in E \quad \text{where} \quad u, v \in V \quad \text{and} \quad u \rightarrow v; \quad (3.1)$$

- $\sigma : E \rightarrow \{C, L\}$ labels every branch as either *capacitive* (C) or *inductive* (L).

We assume G is connected and well-posed. We write $E_C := \sigma^{-1}(C)$ and $E_L := \sigma^{-1}(L)$.

Remark 3.2 (Postponing the naming of a ground). No ground node has been singled out yet, and the omission is a very deliberate one, beyond the mere aesthetic symmetry. The incidence matrix A defined below satisfies $\ker(A^\top) = \text{span}\{|1\rangle\}$ (lemma 3.14). Meaning, any uniform shift produces zero voltage difference across every branch, and is therefore a physically unobservable gauge symmetry.

Prematurely choosing a ground node now, amounts to deleting a row of A (removing a degree of freedom), which resolves this redundancy but at a great cost: all nodes can no longer be treated as equal, singling out the ground as structurally special. Breaking this node permutation symmetry hinders the dual graph construction. Although the same results follow, it requires a careful tracking and propagation of the asymmetry. Instead, we will carry the full $|V|$ -dimensional flux space throughout, and let the kernel of our matrix objects reveal all the null directions simultaneously.

3.1.2. Step 2: Embedding on the Sphere and the Dual Graph

A connected planar graph G may be embedded in the plane $\mathbb{R}^2$, but it is more natural to work on the two-sphere S^2 . The embedding $G \rightarrow S^2$ partitions the surface into faces (open discs). Another reason for working on S^2 rather than $\mathbb{R}^2$ is that the unbounded region of the plane also becomes a face on equal footing with all others, avoiding the asymmetry of treating an “infinite” region any differently. As will be a motif throughout this discussion, symmetry is our best friend.

Definition 3.3 (Faces of a Planar Embedding). Given a planar embedding of G on S^2 , the complement $S^2 \setminus G$ is a disjoint union of open discs called the faces of the embedding, F .

Remark 3.4 (Non-uniqueness of the embedding). A planar graph does not admit a unique embedding in S^2 . Distinct embeddings may exist and partition the sphere into different face sets. All subsequent results hold for a given fixed embedding; in other words, symplectic geometry and dynamical system properties are all embedding-agnostic.

Definition 3.5 (Dual Graph). The dual graph $G^* = (V^*, E^*, \sigma^*)$ of a planar embedding is constructed:

- $V^* = F$: one dual vertex per face of the embedding;
- For each edge $e \in E$ separating faces $f_i \neq f_j$, there is a dual edge $e^* \in E^*$ connecting f_i to f_j , oriented to cross e from right to left (chosen by convention);
- $\sigma^*(e^*)$ is the *opposite* label of $\sigma(e)$: the dual of a capacitive branch is inductive, and vice versa.

The exchange $\sigma^*(e^*) \neq \sigma(e)$ is the key feature of circuit duality. We will see that the constitutive relations of capacitors and inductors are related by exactly swapping the roles of flux and charge, so a circuit and its dual obey the same equations of motion with only the roles of voltage and current interchanged. This fact later motivates the symmetric formulation M .

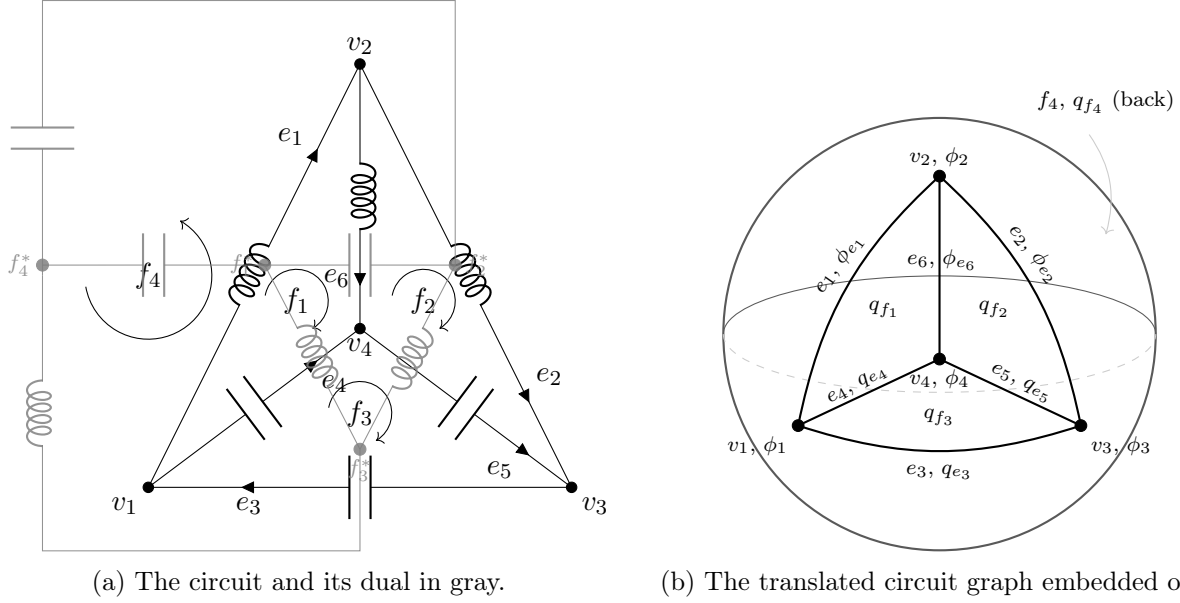


Figure 1: (a) A circuit with four faces, six edges, and four vertices, shown in black; its exact dual is overlaid in gray. Under duality, each face becomes a dual vertex f_i^* , and each primal edge is crossed by a dual edge whose component type is swapped (capacitor $\leftrightarrow$ inductor). (b) The circuit translated into a graph (K_4). Each mathematical object carries the physical variable that lives on it: node fluxes ϕ_v , branch fluxes ϕ_e (drawn only on inductive edges), branch charges q_e (drawn on capacitive edges), and face charges q_f . This example is an expanded analysis of Fig. 2 in [7].

Remark 3.6 (Graph duality vs. electromagnetic duality). It is worth distinguishing two notions of duality at play. The *graph dual* of [definition 3.5](#) acts only on the graph and its C/L labels, swapping σ and σ^* . The *electromagnetic dual* is much stronger: an involution on phase space that swaps $\phi \leftrightarrow q$, capacitors with inductors, and parallel with series. For linear LC circuits, the two notions coincide, since swapping a linear capacitor with a linear inductor is precisely the exchange of ϕ and q in the constitutive relations. For nonlinear elements the two may differ: in the superconducting circuits, relabeling a Josephson Junction ([Section 5.2.1](#)) is not enough, and the existence of the QPS element ([Section 5.2.2](#)) as its electromagnetic dual is a separate physical input. In general, producing a physical circuit realizing the electromagnetic dual is not always possible—however this does not obstruct any of the results or the intuitive discussion of dual that follows.

Definition 3.7 (Incidence Matrix). The incidence matrix $A \in \mathbb{R}^{|V| \times |E|}$ is defined by

$$A_{v,e} := \begin{cases} +1 & \text{if branch } e \text{ ends at node } v, \\ -1 & \text{if branch } e \text{ starts at node } v, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

for $e \in E$ and $v \in V$. We also write A_C and A_L for the submatrices of A corresponding to columns

in E_C and E_L respectively, so that

$$A = \begin{pmatrix} A_C & A_L \end{pmatrix}. \quad (3.3)$$

In the K_4 example of [Figure 1](#), the incidence matrix A would be as follows:

$$A : \mathbb{R}^6 \rightarrow \mathbb{R}^4, \quad A_{v_i e_j} = \begin{array}{c} \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ v_1 & \begin{pmatrix} -1 & 0 & +1 & -1 & 0 & 0 \\ +1 & -1 & 0 & 0 & 0 & -1 \\ 0 & +1 & -1 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 & -1 & +1 \end{pmatrix} \end{matrix} \end{array} \quad (3.4)$$

3.1.3. Step 3: Variables on Faces and Vertices

1. At each node $v \in V$, we examine its node flux $\phi_v \in \mathbb{R}$, defined as follows:

Definition 3.8 (Node Flux).

$$\phi_v(\tau) := \int_{-\infty}^{\tau} V_v(s) ds, \quad \dot{\phi}_v = V_v. \quad (3.5)$$

where V_v is the voltage at a given node. We collect the node fluxes into the single ket $|\phi_V\rangle \in \mathbb{R}^{|V|}$ with components ϕ_v .

2. Along each edge $e \in E$, we examine the branch charge q_e and branch flux ϕ_e . Since every branch carries a current $I_e(\tau)$ and a voltage $V_e(\tau)$, we can take their time integrals to define:

Definition 3.9 (Branch Charge and Flux, Loop Charge).

$$q_e(\tau) := \int_{-\infty}^{\tau} I_e(s) ds, \quad \phi_e(\tau) := \int_{-\infty}^{\tau} V_e(s) ds, \quad (3.6)$$

so $\dot{q}_e = I_e$ and $\dot{\phi}_e = V_e$.

Branch charges are a natural objects that live on individual edges, but for the later symmetric formulation of [Section 6](#), we will require the related object: *loop charge*. In [Section 3.2](#), we will introduce Kirchhoff's Current Law, which states that the algebraic sum of currents at every node is zero. The only way for a current to respect this conservation law is to circulate along a closed path. In a linear algebraic sense, the space of currents that obey this non-accumulation is a linear subspace of $\mathbb{R}^{|E|}$ spanned precisely by the loop vectors of the graph ([lemma 6.5](#)). The *loop current* I_l is then the current circulating around a loop l , and its time integral

$$q_l(\tau) := \int_{-\infty}^{\tau} I_l(s) ds \quad (3.7)$$

is the *loop charge* on l . Note that branch charges are recovered from loop charges by summing the loop charges of all loops passing through the edge, with appropriate orientation signs.

3.2. Kirchhoff's Laws and Constitutive Relations

By now, we have introduced the graph-theoretic representation of a circuit, together with its associated algebraic structure, such as the incidence matrix. We then introduced the physical measurables of flux and charge that live along the graph structure.

The eventual goal is to formulate the equations of motion governing the circuit in terms of Kirchhoff's laws together with the constitutive relations of circuit elements. These relations are stated in this section in a form that is sufficiently general to accommodate both linear and nonlinear components.

Axiom 3.10 (Kirchhoff's Voltage Law (KVL)). *The branch voltage vector lies in $\text{Im}(A^\top)$:*

$$\dot{\phi}_e = (A^\top |\dot{\phi}_V\rangle)_e \quad \text{for all } e \in E. \quad (3.8)$$

Equivalently, the signed sum of branch voltages around any closed loop is zero. Notice that this constraint means branch flux can always be expressed in terms of node fluxes.

Remark 3.11. The incidence matrix A and the dual co-incidence matrix $A^* \in \mathbb{R}^{|F| \times |E|}$ satisfy the exactness relation from algebraic topology:

$$\ker(A^*) = \text{Im}(A^\top). \quad (3.9)$$

This actually encodes KVL: a vector of branch voltages (null space of A^*) that sums to zero around every face is expressible as differences of node potentials (the image of $A^\top$). The proof of the following is analogous to that of the later [lemma 6.5](#).

Axiom 3.12 (Kirchhoff's Current Law (KCL)). *At every node, the algebraic sum of currents vanishes. Geometrically, this means the current vector lies in $\ker(A)$:*

$$A|I\rangle = |0\rangle, \quad |I\rangle = (I_e)_{e \in E}. \quad (3.10)$$

Axiom 3.13 (Constitutive Relations). *Each circuit element imposes a local relation between the branch flux and branch charge on its edge, encoded by a smooth function $f_e : \mathbb{R} \rightarrow \mathbb{R}$:*

(a) *Capacitive branch $e \in E_C$:*

$$\dot{\phi}_e = f_e(q_e). \quad (3.11)$$

The voltage across the branch is some function of the charge stored in it.

(b) *Inductive branch $e \in E_L$:*

$$\dot{q}_e = f_e(\phi_e). \quad (3.12)$$

The current through the branch is some function of the flux threading it.

This formulation is purposefully general to any smooth function, allowing its application in both simple linear constitutive cases, as well as nonlinear circuit elements in the following sections. In the simplest case, linear capacitors have $f_e(q_e) = q_e/C_e$ and linear inductors have $f_e(\phi_e) = \phi_e/L_e$. Note that this arrangement also demonstrates how duality operates by swapping the roles of flux and charge while leaving the functional form of f_e unchanged.

The axioms of KVL, KCL, and the constitutive relations determine the full set of equations of motion for the circuit. However, several lemmas allow us to reformulate these relations in a way that makes their underlying algebraic structure transparent.

Lemma 3.14 (Constant vectors in $\ker(A^\top)$). *For any scalar c , the uniform ket $c|1\rangle$ (all components equal to c) lies in $\ker(A^\top)$. In fact given G is connected, $\ker(A^\top) = \text{span}\{|1\rangle\}$.*

Proof. For any edge $e = (u \rightarrow v)$, $(A^\top |x\rangle)_e = x_v - x_u$. So $A^\top |x\rangle = |0\rangle$ if and only if $|x\rangle$ is constant on every connected component of G . Trivially, the constants are spanned by $|1\rangle$. $\square$

Lemma 3.15 (Kirchhoff's laws in face-node coordinates). *Let $G = (V, E, \sigma)$ be a connected planar circuit graph embedded on S^2 with face set F . Assign node fluxes $|\phi_V\rangle \in \mathbb{R}^{|V|}$ and face charges $|q_F\rangle \in \mathbb{R}^{|F|}$, and define branch fluxes $\phi_e := (A^\top |\phi_V\rangle)_e$ and branch charges $q_e := (A^{*\top} |q_F\rangle)_e$. Then:*

- (i) (KVL) The signed sum of branch voltages around every face $f \in F$ vanishes if and only if $|\phi_E\rangle \in \text{Im}(A^\top)$, which holds automatically for any $|\phi_V\rangle \in \mathbb{R}^{|V|}$ by definition.
- (ii) (KCL) The signed sum of currents at every node $v \in V$ vanishes if and only if $|I\rangle \in \ker(A)$, which holds automatically for any $|q_F\rangle \in \mathbb{R}^{|F|}$ via $I_e = \dot{q}_e = (A^{*\top}|\dot{q}_F\rangle)_e$.

Consequently, in $(|\phi_V\rangle, |q_F\rangle)$ coordinates, Kirchhoff's laws take the compact form

$$\sum_{v \in V} \dot{\phi}_v = \sum_{f \in F} \dot{q}_f = 0. \quad (3.13)$$

Proof. Parts (i) and (ii) are immediate from the exactness of the chain complex of G ((3.9) and its dual): the defining substitutions $|\phi_E\rangle := A^\top |\phi_V\rangle$ and $|I\rangle = A^{*\top} |\dot{q}_F\rangle$ place the branch-flux and branch-current vectors in $\text{Im}(A^\top) = \ker(A^*)$ and $\text{Im}(A^{*\top}) = \ker(A)$ respectively, so KVL and KCL are satisfied automatically.

The identities (3.13) are then consequences of being gauge symmetries. By lemma 3.14, $|1\rangle \in \ker(A^\top)$: the branch fluxes $\phi_e = (A^\top |\phi_V\rangle)_e$ are invariant under the uniform shift $|\phi_V\rangle \mapsto |\phi_V\rangle + c|1\rangle$, a pure gauge. A representative of each gauge orbit thus $\sum_v \phi_v = 0$.

Differentiating in time gives the first identity $\sum_v \dot{\phi}_v = 0$. The dual argument applied to $|1\rangle \in \ker(A^{*\top})$ singles out the uniform shift of face charges as gauge and yields the second identity $\sum_f \dot{q}_f = 0$. Intuitively, being identically 0 means nothing in the dynamical system can couple to it, thus carrying no physical information (hence being pure gauges).

Both identities are, therefore, automatic consequences of our graph structure. A more direct and explicit proof of the same facts, organized around the column-sum identity $A|1\rangle = |0\rangle$ and its dual, is provided in Section A.1. $\square$

Remark 3.16 (Faces vs. loops). Throughout this paper we restrict to planar circuits ([7] extends to non-planar). In this setting, “face charges” and “loop charges” refer to the same underlying degrees of freedom. From a graph-theoretic standpoint, a face is an enclosed region of the embedding (no crossing edges), while a loop is any closed walk in the graph. For a planar embedding on S^2 , every face boundary is a loop, and every loop decomposes as a signed sum of face boundaries; rigorously, the face boundaries span the cycle space of G .

This spanning set is intentionally redundant: the $|F|$ face boundaries satisfy a single global relation $\sum_{f \in F} B_{f,e} = 0$ (every edge borders exactly two faces with opposite traversal), so they span an $(|E| - |V| + 1)$ -dimensional cycle space with one degree of redundancy, expressed equivalently as $\sum_{f \in F} \dot{q}_f = 0$ from lemma 3.15. Keeping all $|F|$ faces—including the external outer face on S^2 —is faithful to the embedding, treats no face as privileged (which would be prematurely fixing a gauge), and makes the flux-charge symmetry (definition 6.7) and the Euler characteristic visible in the degree-of-freedom count (corollary 7.6) later. We therefore use $f \in F$ throughout to index loop coordinates, and reserve the word “loop” for elements of the cycle space (closed walks in G , including homogeneous loops and fundamental cycles).

Now, we have seen a full translation of a physical circuit object into a purely mathematical one. This graph-theoretic formulation provides a transparent view of the circuit, with physical constraints cooked into its very topology and a system of equations.

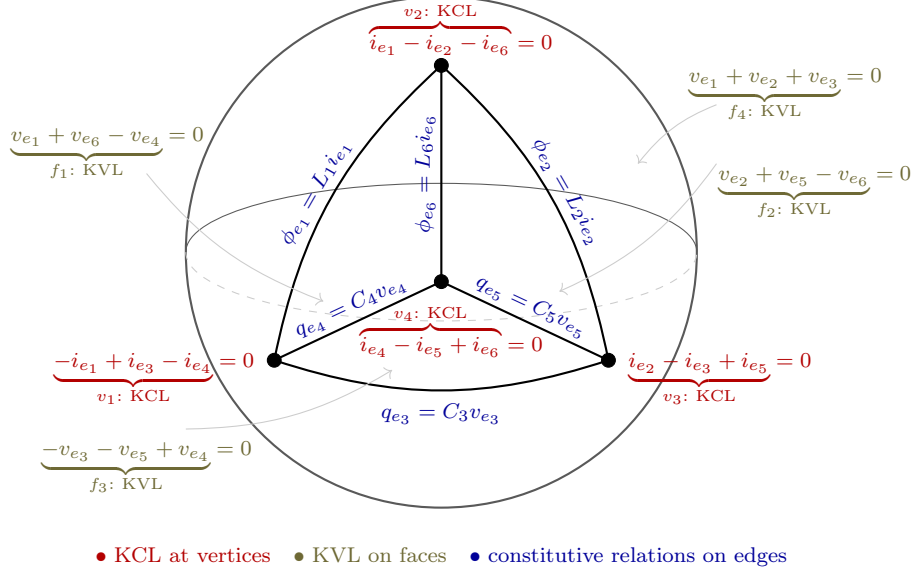


Figure 2: The graph-theoretic system of equations for the circuit of Figure 1, laid out on the embedded graph. KVL lives on faces, KCL lives on vertices, while constitutive relations live on their corresponding edge.

3.3. The Circuit Equations of Motion

We now derive the equations of motion (EOM) for a general well-posed LC circuit graph $G = (V, E, \sigma)$ directly from the physical axioms. We avoid assigning any constitutive relation functions f_e , meaning the derived EOM applies to all circuits with smooth inductive and capacitive components. The graph-theoretic framework now pays off, by working in coordinates $(|\phi_V\rangle, |q_F\rangle)$, Kirchhoff's laws are absorbed into the coordinate choice, leaving the constitutive relations as the only additional content.

Foreshadowing slightly, once we construct a symmetric circuit action S , there is a particularly clean interpretation that: $\delta S / \delta \phi_v = 0$ is precisely Kirchhoff's Current Law at node v , and $\delta S / \delta q_f = 0$ is precisely Kirchhoff's Voltage Law around face f . The Euler-Lagrange equations then are precisely Kirchhoff's laws in the two natural coordinates, and correspond to the equation of motion derived in this section. This will be proven explicitly in theorem 6.10.

Definition 3.17 (The Circuit Equation of Motion). The natural coordinate space for this system is $(|\phi_V\rangle, |q_F\rangle) \in \mathbb{R}^{|V|} \times \mathbb{R}^{|F|}$, where $|\phi_V\rangle$ collects node fluxes and $|q_F\rangle$ collects face charges.

First, recall that KVL gives the branch flux on any edge directly as a difference of node fluxes:

$$\phi_e = (A^\top |\phi_V\rangle)_e \quad \text{for all } e \in E. \quad (3.14)$$

Symmetrically, lemma 3.15 expresses each branch charge as a signed sum of face charges, $q_e = (A^* |\q_F\rangle)_e$. Therefore, any smooth trajectory $(|\phi_V\rangle(\tau), |q_F\rangle(\tau))$ automatically encodes a physically valid configuration for Kirchhoff's Laws.

For each branch $e \in E$, let $f_e : \mathbb{R} \rightarrow \mathbb{R}$ be the smooth function encoding the constitutive relation of that element. Define

$$\Sigma_e := \begin{cases} \dot{\phi}_e - f_e(q_e) & e \in E_C, \\ \dot{q}_e - f_e(\phi_e) & e \in E_L. \end{cases} \quad (3.15)$$

Together with the global Kirchhoff redundancies from [lemma 3.15](#),

$$\Lambda_1 := \sum_{v \in V} \dot{\phi}_v, \quad \Lambda_2 := \sum_{f \in F} \dot{q}_f, \quad (3.16)$$

the circuit dynamical system arising from G takes on the concise equations of motion:

$$\boxed{\Sigma_e = \Lambda_1 = \Lambda_2 = 0 \quad \forall e \in E.} \quad (3.17)$$

Remark 3.18 (Linear circuits and the weighted Laplacian form). When all constitutive relations are linear ($f_e(x) = x/C_e$ for capacitive, $f_e(x) = x/L_e$ for inductive), the general system (3.17) reduces to a single closed second-order ODE in node flux. Define the two weighted graph Laplacians

$$C := A_C C_b A_C^\top \in \mathbb{R}^{|V| \times |V|}, \quad K := A_L L_b^{-1} A_L^\top \in \mathbb{R}^{|V| \times |V|}, \quad (3.18)$$

where $C_b = \text{diag}(C_e)_{e \in E_C}$ and $L_b = \text{diag}(L_e)_{e \in E_L}$. A bit of manipulation [1] (including assumptions made in [theorem 4.3](#) and isolating for $\dot{q}_C$ and $\dot{q}_L$) reveals that the circuit equation of motion takes on the familiar harmonic oscillator form of

$$C |\ddot{\phi}_V\rangle + K |\phi_V\rangle = |0\rangle. \quad (3.19)$$

The proof of which is omitted as it is not the focus of this paper and is standard in the literature.

Lemma 3.19 (Properties of C and K). *Both C and K defined in [remark 3.18](#) are symmetric and positive semi-definite.*

Proof. Symmetry: $(A_C C_b A_C^\top)^\top = A_C C_b^\top A_C^\top = A_C C_b A_C^\top$ since C_b is diagonal. The analogous statement holds for K .

Positive semi-definiteness: For any $|x\rangle \in \mathbb{R}^{|V|}$,

$$\langle x | C | x \rangle = \langle x | A_C C_b A_C^\top | x \rangle = \sum_{e \in E_C} C_e [(A_C^\top | x \rangle)_e]^2 \geq 0,$$

since every $C_e > 0$. The analogous statement holds for K with $L_e^{-1} > 0$. $\square$

Remark 3.20 (A counting argument via Euler's formula). The algorithm involves $|V| + |F|$ unknowns (ϕ_v, q_f) and $|E|$ constitutive relations. Euler's formula for a connected planar graph on S^2 ,

$$|V| - |E| + |F| = 2,$$

provides a remarkable insight. The constitutive relations underdetermine the system by exactly two equations—precisely the two Kirchhoff constraints $\Lambda_1 = \Lambda_2 = 0$. By fixing the choice of coordinates and thus unknowns, one can see this counting property in [Figure 2](#). As a reminder, the Kirchhoff law equations are not all independent and can be related via [lemma 3.15](#). It is no coincidence that Euler's formula, a purely topological fact, forces the system to close. We will return to this remark after the final theorem in [Section 9.2](#).

4. LINEAR CIRCUIT DYNAMICAL SYSTEMS ARE HAMILTONIAN

We now use the circuit dynamical system constructed in [Section 3](#), specifically the equation of motion (3.19) together with the weighted Laplacians C and K from (3.18), to show that any finite *linear* LC circuit is a Hamiltonian dynamical system. The central difficulty is that C is in general *singular*, which means an immediate Legendre transform fails and we must first reduce to a quotient space.

As the discussion progresses, we will see that singularity is a central concern of relating dynamical systems—it is through identifying the null directions that all the major results proceed.

4.1. The Poisson Bracket

The bracket that will eventually serve as our Hamiltonian structure will seem to be pulled from thin air, conveniently satisfying all constraints. The purpose of this precursor section is to motivate that the Poisson bracket is the *unique* bracket compatible with the symplectic structure of the underlying Lagrangian—both the linear first-order form (4.4) introduced in the next section, and the more general symmetric circuit action (6.9). A rigorous derivation is provided in Section A.2.

Definition 4.1 (Poisson Bracket). For smooth functions $f, g \in C^\infty(P)$ on a phase space $P = \mathcal{V} \oplus \mathcal{V}^*$, the Poisson bracket is:

$$\{f, g\}(|\phi_{(\cdot)}\rangle, |q_{(\cdot)}\rangle) := \langle \nabla_\phi f | \nabla_q g \rangle - \langle \nabla_q f | \nabla_\phi g \rangle, \quad (4.1)$$

where ∇_ϕ and ∇_q denote gradients in the $\mathcal{V}$ and $\mathcal{V}^*$ directions respectively.¹

Lemma 4.2 (Properties of the Bracket). *If P is a reduced phase space (all null directions projected out), the bracket (4.1) is antisymmetric, non-degenerate, and bilinear, and therefore satisfies the requirements of definition 2.9.*

Proof. Each property proceeds separately:

1. *Antisymmetry:* $\{f, g\} = \langle \nabla_\phi f | \nabla_q g \rangle - \langle \nabla_q f | \nabla_\phi g \rangle = -\{g, f\}$. This antisymmetry is forced by the requirement that $(|\phi_{(\cdot)}\rangle, |q_{(\cdot)}\rangle)$ be canonically conjugate: a symmetric bracket would make conjugacy impossible.
2. *Non-degeneracy:* Suppose $\{f, g\} = 0$ for all f . Taking $f = \phi_i$ gives $\partial g / \partial q_i = 0$ for all i ; taking $f = q_i$ gives $\partial g / \partial \phi_i = 0$ for all i . Therefore $\nabla g \equiv 0$ on P , and since all null directions have been removed, $g = 0$.
3. *Bilinearity:* Immediate from linearity of the gradient and bilinearity of the inner product. $\square$

A direct computation confirms the canonical relations:

$$\{\phi_i, \phi_j\} = 0, \quad \{q_i, q_j\} = 0, \quad \{\phi_i, q_j\} = \delta_{ij}. \quad (4.2)$$

4.2. Linear Circuits are Hamiltonian

We are now equipped with all the necessary tools to relate a circuit dynamical system to a Hamiltonian one under the assumption that all constitutive relations are linear.

A brief note on the choice of charge variable. The natural phase space coordinates from definition 3.17 are $(|\phi_V\rangle, |q_F\rangle)$ (node fluxes and face charges). In the linear setting, the constitutive relation $q_e = C_e \phi_e$ is invertible, meaning charge can be eliminated algebraically in favor of ϕ_e , and the configuration space reduces to $|\phi_V\rangle$ alone. The Legendre transform produces a conjugate momentum on the dual node space $\mathcal{V}^*$, which we denote $|q_V\rangle$. This substitution is unavailable later in the nonlinear case where we symmetrically treat charge as an independent variable from the outset and return to the idea of face/loop charges.

¹Both (4.4) and (6.9) omit the factor of $\frac{1}{2}$ that often appears with the symplectic term in the Lagrangian. This is a normalization convention (it gets absorbed into $H = \frac{1}{2}E$ and ω^{-1} upon passing to the bracket).

Theorem 4.3. *Any finite linear circuit dynamical system is a Hamiltonian dynamical system.*

More precisely, let $G = (V, E, \sigma)$ be a finite well-posed circuit graph with linear constitutive relations. Then the circuit equation of motion (3.19) is a Hamiltonian dynamical system on a reduced phase space $\mathcal{V} \oplus \mathcal{V}^*$.

Proof. We proceed in six steps, working with the objects from Section 3: the node flux ket $|\phi_V\rangle \in \mathbb{R}^{|V|}$, the weighted Laplacians $C = A_C C_b A_C^\top$ and $K = A_L L_b^{-1} A_L^\top$, the Poisson bracket (definition 4.1), and the equation of motion (3.19).

Step 1: The Equation of Motion is a Linear Dynamical System

The equation of motion (3.19) is a second-order linear ODE, so it is a linear dynamical system by definition 2.3. We take the following quadratic Lagrangian by ansatz (saving the longer constructive discussion for the general case):

$$L(|\phi_V\rangle, |\dot{\phi}_V\rangle) = \frac{1}{2} \langle \dot{\phi}_V | C | \dot{\phi}_V \rangle - \frac{1}{2} \langle \phi_V | K | \phi_V \rangle, \quad (4.3)$$

whose Euler-Lagrange equations reproduce (3.19) by symmetry of C and K . The two terms are precisely the total capacitive and inductive energies:

$$\frac{1}{2} \langle \dot{\phi}_V | C | \dot{\phi}_V \rangle = \sum_{e \in E_C} \frac{1}{2} C_e V_e^2 = E_C, \quad \frac{1}{2} \langle \phi_V | K | \phi_V \rangle = \sum_{e \in E_L} \frac{1}{2} L_e^{-1} \phi_e^2 = E_L,$$

Equivalently, getting slightly ahead of ourselves, introducing $|q_V\rangle \in \mathcal{V}^*$ as an independent variable conjugate to $|\phi_V\rangle$, the same dynamics arise from the first-order Lagrangian

$$L_{\text{sym}}(|\phi_V\rangle, |q_V\rangle, |\dot{\phi}_V\rangle) = \langle q_V | \dot{\phi}_V \rangle - H(|\phi_V\rangle, |q_V\rangle), \quad (4.4)$$

on the reduced phase space given that C is invertible (which it currently is not). Varying $|q_V\rangle$ recovers $|q_V\rangle = C|\dot{\phi}_V\rangle$, and substituting back reproduces (4.3); varying $|\phi_V\rangle$ gives (3.19) directly. The symplectic term $\langle q_V | \dot{\phi}_V \rangle$ has the form $\omega_{\alpha\beta} x^\alpha \dot{x}^\beta$ of (A.1) with the block-identity ω , making the symplectic structure manifest at the level of the Lagrangian. We include this form to demonstrate the connection to the symplectic discussion of the Poisson Bracket Section 4.1 and to motivate the subsequent steps that resolve the general invertibility of C .

Step 2: The Null Space of C and Its Graph-Theoretic Meaning

Legendre transform requires C to be invertible, but $C = A_C C_b A_C^\top$ does not guarantee this. We first understand what each null direction represents.

Lemma 4.4 (Null space of C). *$\ker C = \ker(A_C^\top)$, and is spanned by the indicator vectors of the connected components of the capacitive subgraph (V, E_C) . In particular, $\dim(\ker C) = k$ where k denotes the number of capacitively connected components.*

Proof. By assumption, C_b is diagonal with strictly positive entries, meaning

$$A_C C_b A_C^\top |x\rangle = |0\rangle \iff A_C^\top |x\rangle = |0\rangle,$$

giving $\ker C = \ker(A_C^\top)$. Applying lemma 3.14 to (V, E_C) completes the proof. $\square$

Any uniform shift on a connected component of (V, E_C) leaves all branch voltages unchanged. This is the gauge freedom of choosing a ground node per capacitive component.

Step 3: Reduction to the Gauge-Fixed Quotient Space

Define the *reduced flux space* $\mathcal{V} := \mathbb{R}^{|V|} / \ker C$, a finite-dimensional real vector space of dimension $n := |V| - k$. Fixing a representative in $\mathcal{V}$ amounts to fixing a ground node per component.

Lemma 4.5 (Invertibility of C on $\mathcal{V}$). *The restriction $C|_{\mathcal{V}} : \mathcal{V} \rightarrow \mathcal{V}^*$ is symmetric positive definite, hence invertible.*

Proof. Let $|x\rangle \in \mathcal{V} \setminus \{|0\rangle\}$. Since $\mathcal{V} = (\ker C)^\perp$, we have $|x\rangle \notin \ker C$, so $A_C^\top |x\rangle \neq |0\rangle$ by lemma 4.4. Therefore, by positive definiteness of C_b :

$$\langle x|C|x\rangle = \langle x|A_C C_b A_C^\top |x\rangle = \sum_{e \in E_C} C_e [(A_C^\top |x\rangle)_e]^2 > 0.$$

Symmetry follows from lemma 3.19. Positive definiteness implies invertibility. $\square$

Step 4: First-Order Form on the Reduced Phase Space

Define the *conjugate momentum*

$$|q_V\rangle := C|\dot{\phi}_V\rangle \in \mathcal{V}^*. \quad (4.5)$$

Since $C|_{\mathcal{V}}$ is invertible, $|\dot{\phi}_V\rangle = C^{-1}|q_V\rangle$. The equation of motion then becomes the first-order system:

$$|\dot{\phi}_V\rangle = C^{-1}|q_V\rangle, \quad |\dot{q}_V\rangle = -K|\phi_V\rangle. \quad (4.6)$$

Step 5: The Poisson Bracket on $P = \mathcal{V} \oplus \mathcal{V}^$*

The canonical Poisson bracket (4.1) was verified to be antisymmetric, non-degenerate, and bilinear in lemma 4.2. The canonical relations (4.2) confirm that $(|\phi_V\rangle, |q_V\rangle)$ are conjugate coordinates on P .

Step 6: The Hamiltonian and Verification of Hamilton's Equations

Define the Hamiltonian as the total stored energy:

$$H(|\phi_V\rangle, |q_V\rangle) = \underbrace{\frac{1}{2}\langle q_V|C^{-1}|q_V\rangle}_{\text{capacitive energy}} + \underbrace{\frac{1}{2}\langle \phi_V|K|\phi_V\rangle}_{\text{inductive energy}}. \quad (4.7)$$

This is the Legendre transform of (4.3) on $\mathcal{V}$. Computing gradients, $\nabla_q H = C^{-1}|q_V\rangle$ and $\nabla_\phi H = K|\phi_V\rangle$. Hamilton's equations then give:

$$|\dot{\phi}_V\rangle = \nabla_q H = C^{-1}|q_V\rangle, \quad |\dot{q}_V\rangle = -\nabla_\phi H = -K|\phi_V\rangle, \quad (4.8)$$

recovering (4.6) and hence (3.19). All conditions of definition 2.9 are satisfied. $\square$

5. THE NON-LINEARITY CONUNDRUM

5.1. Taking Inventory

So far, we have specified how to translate a finite planar circuit into a mathematical object: a graph embedded on S^2 with flux and charge as natural coordinates. When the circuit dynamical system is linear, the weighted Laplacians C and K encode topological and constitutive information, and a well-defined Hamiltonian can be constructed after quotienting out the null directions in $\ker C$.

The canonical pair $(|\phi_V\rangle, |q_V\rangle)$ represents node fluxes and total charges accumulated at each node, satisfying $\{\phi_i, q_j\} = \delta_{ij}$. Foreshadowing, in the nonlinear case, flux and charge will still be the canonical physical quantities, but redefined on branches and loops. In quantum mechanics, promoting these to operators gives the Schrödinger equation and the energy spectrum of the circuit.

5.2. Superconducting Circuits

A superconducting circuit operates at extremely low temperatures, where quantum effects dominate. By promoting (ϕ, q) to operators, the quantized Hamiltonian governs the energy spectrum, and the system can be engineered to have only two accessible low-energy eigenstates—a qubit.

Consider a capacitor in parallel with a Josephson Junction (a Transmon circuit). A Josephson Junction consists of two superconductors separated by a thin insulating barrier through which Cooper pairs tunnel coherently. This gives rise to a nonlinear inductive element whose energy is sinusoidal, introducing anharmonicity into the spectrum and enabling qubit isolation [3].

Therefore, it is particularly valuable to be able to characterize superconducting circuit dynamical systems, and it is a natural next step to orient our discussion in this direction.

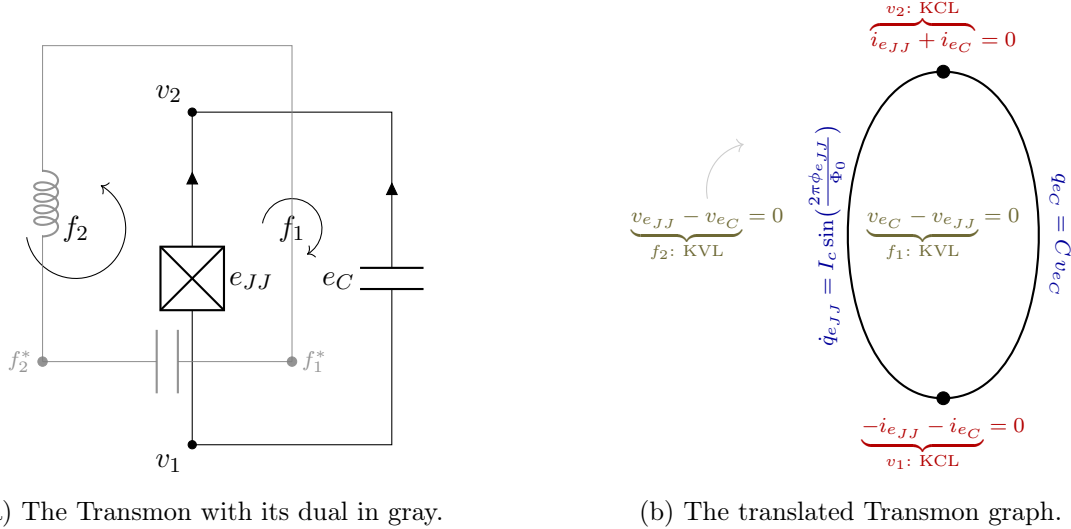


Figure 3: (a) The Transmon: a Josephson Junction e_{JJ} in parallel with a shunt capacitor e_C , with $V = \{v_1, v_2\}$, $E_L = \{e_{JJ}\}$, $E_C = \{e_C\}$; its dual is overlaid in gray. (b) The translated graph, with KCL at each vertex, KVL on each face, and the constitutive relation displayed on each edge. The two KCL equations and the two KVL equations are linearly dependent (one of each is redundant).

5.2.1. The Josephson Junction Constitutive Relation

In order to analyze superconducting circuits, we must extend our framework to include new constitutive relations corresponding to additional circuit elements. Among these, the Josephson junction plays a central role. Unlike linear inductive or capacitive elements, it introduces an intrinsically nonlinear relation between flux and current, reflecting its quantum mechanical origin. We therefore begin by stating its constitutive relation as it fits into our general formalism (Axiom 3.13).

Axiom 5.1 (Josephson Junction Constitutive Relation). *Given a JJ branch $e \in E_{JJ} \subseteq E_L$, where $I_c > 0$ is the critical current and $\Phi_0 = h/(2e)$ is the superconducting flux quantum, the constitutive*

relation takes the form of (3.12) with

$$f_e(\phi_e) = I_c \sin\left(\frac{2\pi \phi_e}{\Phi_0}\right), \quad \text{so} \quad \dot{q}_e = I_c \sin\left(\frac{2\pi \phi_e}{\Phi_0}\right). \quad (5.1)$$

The energy stored in a JJ branch is:

$$E_{JJ}(\phi_e) = -E_J \cos\left(\frac{2\pi \phi_e}{\Phi_0}\right), \quad E_J := \frac{\Phi_0 I_c}{2\pi}. \quad (5.2)$$

Remark 5.2 (The Transmon and the limits of the linear proof). The Transmon is a Josephson Junction in parallel with a shunt capacitor C (Figure 3). The equation of motion becomes

$$C \ddot{\phi} + I_c \sin\left(\frac{2\pi \phi}{\Phi_0}\right) = 0,$$

a nonlinear pendulum in the gauge-fixed coordinate $\phi = \phi_1 - \phi_2$. Although the six-step proof does not apply directly (since f_e is not linear), the Transmon can still be quantized by other methods—the Lagrangian is well-defined and the Legendre transform can be carried through. The Hamiltonian $H = q^2/2C - E_J \cos(2\pi\phi/\Phi_0)$ is anharmonic and yields a well-defined two-level system [3]. The Transmon is not a meaningful obstruction. A circuit that resists the approach is the Dualmon.

5.2.2. The Dualmon

The central example we will use to highlight superconducting nonlinear circuit dynamics is the Dualmon, an extension of the dual of a Transmon. Recall that the dual graph swaps capacitive and inductive branches; accordingly, the Transmon's Josephson Junction (heuristically a nonlinear periodic inductor) dualizes to a *quantum phase slip* (QPS) element (heuristically a capacitor).

A QPS element is a capacitive superconducting nanowire in which Cooper pairs undergo coherent quantum tunneling of magnetic flux, long story short, producing a constitutive relation that is the flux-charge dual of the JJ.

Axiom 5.3 (Quantum Phase Slip Constitutive Relation). *Given a QPS branch $e \in E_{QPS}$, where $V_Q > 0$ is the phase slip voltage amplitude and $2e$ is the Cooper pair charge, the constitutive relation takes the form of a capacitive element (3.11) with*

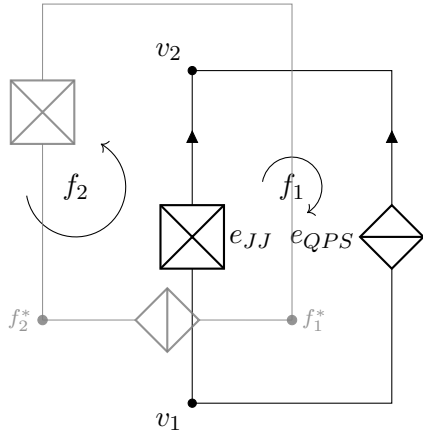
$$f_e(q_e) = V_Q \sin\left(\frac{2\pi q_e}{2e}\right), \quad \text{so} \quad \dot{\phi}_e = V_Q \sin\left(\frac{2\pi q_e}{2e}\right). \quad (5.3)$$

The energy stored in a QPS branch is:

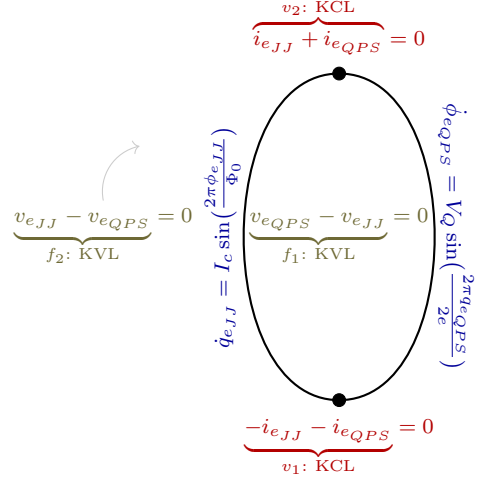
$$E_{QPS}(q_e) = -E_Q \cos\left(\frac{2\pi q_e}{2e}\right), \quad E_Q := \frac{2e V_Q}{2\pi}. \quad (5.4)$$

The Dualmon circuit takes the QPS element (e_{QPS} , capacitive) and goes one step further, placing it in series with a Josephson Junction (e_{JJ} , inductive), forming a loop between nodes v_1 and v_2 (Figure 4). Therefore, while the Transmon was only periodic in flux, the Dualmon is periodic in both flux and charge (double periodicity), giving it special spectrum properties [4]. The dual of the dualmon has the same circuit structure, a fact reflected in its name.

$$V = \{v_1, v_2\}, \quad E_C = \{e_{QPS}\}, \quad E_L = \{e_{JJ}\}, \quad A_C = A_L = \begin{pmatrix} +1 \\ -1 \end{pmatrix} \in \mathbb{R}^{2 \times 1}. \quad (5.5)$$



(a) The Dualmon with its dual in gray.



(b) The translated Dualmon graph.

Figure 4: (a) The Dualmon: a Josephson Junction e_{JJ} in series with a quantum phase slip element e_{QPS} forming a loop between v_1 and v_2 ; its dual is overlaid in gray. (b) The translated graph with its equations of motion. Notice that the underlying graph is identical to that of the Transmon (Figure 3); the Dualmon is distinguished entirely by its constitutive relations, which produce the flux-charge symmetric structure that resists the linear quantization proof.

Algorithm 5.4 (The Dualmon Equation of Motion). Deriving the particular Dualmon equation of motion beyond the general (3.17) follows from three substitutions:

Step 1: By KVL, both branches see the same branch flux:

$$\phi_{e_{JJ}} = \phi_{e_{QPS}} = \phi_1 - \phi_2 =: \phi. \quad (5.6)$$

Step 2: Applying the constitutive relations:

1. QPS branch: $\dot{\phi}_{e_{QPS}} = V_Q \sin(2\pi q_{e_{QPS}}/2e)$.
2. JJ branch: $\dot{q}_{e_{JJ}} = I_c \sin(2\pi\phi/\Phi_0)$.

Step 3: KCL at v_1 gives $\dot{q}_{e_{QPS}} + I_c \sin(2\pi\phi/\Phi_0) = 0$. Together with Step 2, the equations of motion in the gauge-fixed coordinate $\phi = \phi_1 - \phi_2$ are:

$$\dot{\phi} = V_Q \sin\left(\frac{2\pi q}{2e}\right), \quad \dot{q} = -I_c \sin\left(\frac{2\pi\phi}{\Phi_0}\right), \quad \text{where } q := q_{e_{QPS}}. \quad (5.7)$$

5.2.3. Where the Proof Breaks Down

The Dualmon equations of motion (5.7) are perfectly well-posed first-order ODEs in (ϕ, q) . The failure is specifically with the attempt to derive a Hamiltonian via the node-flux Lagrangian approach of Section 4.2.

As a reminder, the existing approach requires writing a Lagrangian purely in node-flux coordinates $|\phi_V\rangle$, followed by performing a Legendre transform. This requires expressing every branch current as a function of $|\phi_V\rangle$, which in turn requires inverting the capacitive constitutive relations to write $q_e = q_e(\dot{\phi}_e)$. For the QPS element, the constitutive relation is $\dot{\phi}_e = V_Q \sin(2\pi q_e/2e)$, giving $\dot{\phi}_e$ as a function of q_e , not the other way around. Unlike the case of a linear capacitor, the sin is not globally invertible, making the inversion ill-defined.

As a result, one cannot write a node-flux Lagrangian in the form $L(|\phi_V\rangle, |\dot{\phi}_V\rangle)$. There simply is nothing to Legendre-transform and our prior approach stops before it can even begin.

A more robust strategy is needed—one that keeps charge as an independent variable from the outset rather than trying to eliminate it. More importantly, we will see that this approach should hold flux and charge on equal footing. The following sections develop this idea.

6. CIRCUIT AND LAGRANGIAN DYNAMICAL SYSTEMS

The breakdown of the linear proof on the Dualmon is not an accident. Despite possessing perfectly well-posed dynamics, the node-flux Lagrangian approach fails because it requires Legendre transform, which in turn requires the capacitance matrix to be non-degenerate, and the Dualmon violates this.

The circuit dynamical system was already stated in full generality in [definition 3.17](#): the conditions $\Sigma_e = \Lambda_1 = \Lambda_2 = 0$ must hold for any smooth constitutive relations f_e . We will show that this general system arises from a single action principle involving node fluxes ϕ_v and loop charges q_f as independent variables. The key object mediating the coupling is not the capacitive incidence matrix $A_C^\top$ —which, as will be explained, is asymmetrically biased toward capacitors—but a new matrix M that treats capacitors and inductors on equal footing. As a spoiler, the canonical pairs that ultimately emerge after gauge reduction will be branch fluxes and an extension of loop charges identified through a spanning tree, of the form (Φ_t, Q_t) .

6.1. Why the Capacitive Incidence Matrix Is Not Enough

It is tempting to write the circuit action by coupling node fluxes to branch charges through the capacitive incidence submatrix $\Omega := A_C^\top$. Such an action *does* reproduce the correct equations of motion, and in fact this is precisely the approach of [\[6\]](#). So why introduce a new matrix at all?

The answer is symmetry. Ω privileges capacitors in a way that nothing about the underlying physics demands. The symplectic pairing $\sum_{e \in E_C} q_e \Omega_{e,v} \dot{\phi}_v$ runs only over capacitive edges, and inductive branches enter only through the energy functional. Under the flux-charge duality, exchanging the capacitors and inductors, it is clear that this Lagrangian does not transform well. The dual theory would pair loop charges with branch fluxes through $A_L^\top$, an entirely different looking object. The two formulations are physically equivalent, but the asymmetry leads to unintuitive mathematical objects that aren't transparent in their structure.

The connection matrix M fixes exactly this. As we will see, M sees capacitive and inductive edges symmetrically through the parity operator P , and the symplectic term $\sum_{f,v} q_f M_{f,v} \dot{\phi}_v$ is structurally invariant under $E_C \leftrightarrow E_L$: swapping the two merely sends $M \mapsto -M$, which exchanges the roles of $|\phi_V\rangle$ and $|q_F\rangle$ in the pairing. This is the flux-charge symmetry of [\[7\]](#), and it is what motivates the entire construction that follows. Beyond aesthetics, this symmetry has notable implications: the obstructions to inverting M are themselves graph-theoretic objects, so the question of whether a circuit admits a physical dual reduces down to a question about algebraic topology.

The practical difference shows up in how simple structural features of a circuit are handled. Consider two capacitors in series, closed into a loop by a Josephson junction ([Figure 5](#)). The middle node v_2 sits between two capacitors, with no inductive path leaving the capacitive chain at that point.

As we will eventually see, in the M -based formulation, the ϕ_{v_2} direction is simply a right null vector of M : meaning the middle node is non-dynamical. Structurally, this is because every loop passing through v_2 must enter via one capacitive edge and leave via the other, so the two contributions cancel exactly at v_2 . The spanning tree construction of [Section 8](#) eliminates this direction automatically

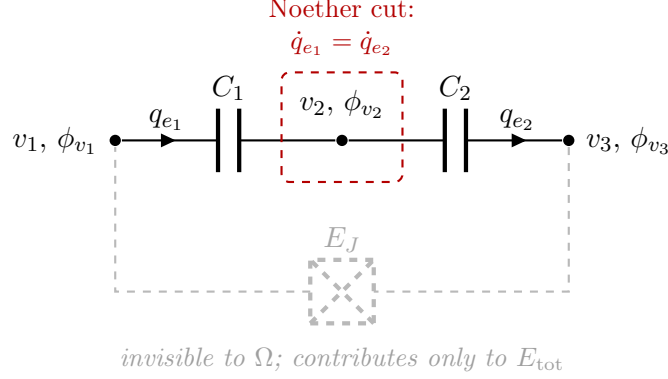


Figure 5: The two-capacitor loop above is two capacitors in series connected in parallel to a Josephson Junction. The symplectic pairing $\sum_{e \in E_C, v} q_e \Omega_{e,v} \dot{\phi}_v$ runs only over the black capacitive edges, meaning the Josephson branch is invisible to Ω and enters the action only through the energy term. Consequently, the non-dynamical direction at the middle node v_2 does not appear as a null vector of Ω , but as an external Noether constraint.

alongside the usual uniform gauge direction. However, in the Ω -based formulation, the same middle node corresponds instead to a *Noether current* [2]—a conservation law (charge conservation across the capacitive cut at v_2) that must be imposed as an extra constraint, and which drags along a “free particle” coordinate that has to be integrated out by hand before the canonical pairs can be identified (Figure 5). Physics aside, the point is Ω seems not to offer a closed system to understand the null directions of the circuit, whereas M does. Both formulations eventually arrive at the same Hamiltonian, but while the Ω -approach requires tracking separate structural features, the M -approach absorbs them all into the machinery.

6.2. The Loop Set and Orientation Matrix

As previously discussed, loop charges are a better object to study than branch charges for two reasons. First, a loop is a cycle in the full graph, so it traverses capacitive and inductive branches symmetrically. Second, the dual exactness $\ker(A) = \text{Im}(A^{*\top})$ from lemma 3.15 already told us that any KCL-respecting current is a signed sum around loops; promoting loop charges as the direct fundamental variable to manipulate therefore encodes KCL at the level of the coordinates.

To capture loop structure, we need a basis for the loop space of the full graph.

Definition 6.1 (Face Set). For a planar circuit graph $G = (V, E, \sigma)$ embedded on S^2 , the face set $F = \{f_1, \dots, f_{|F|}\}$ consists of the connected regions of $S^2 \setminus G$ (the embedding faces, including the outer face). Each face f is assigned an orientation. By Euler’s formula on S^2 , $|F| = |E| - |V| + 2$.

Remark 6.2 (Loops as superpositions of faces). A general *loop* is any closed directed walk in G , not necessarily a face boundary. The cycle space has dimension $|E| - |V| + 1 = |F| - 1$, so the face set F is a redundant spanning set with exactly one relation: the outer face is the signed sum of the inner faces (remark 3.16). One could instead choose a non-redundant basis of $|F| - 1$ loops by deleting one face, but this would single out a face and obscure the symmetric treatment that motivates working on S^2 in the first place.

Definition 6.3 (Orientation Matrix). The orientation matrix $B \in \mathbb{R}^{|F| \times |E|}$ encodes how each face

boundary traverses each edge:

$$B_{f,e} := \begin{cases} +1 & \text{if face boundary } f \text{ and edge } e \text{ are oriented alike,} \\ -1 & \text{if face boundary } f \text{ and edge } e \text{ are oriented opposite,} \\ 0 & \text{if edge } e \text{ does not lie on the boundary of } f. \end{cases} \quad (6.1)$$

The total loop charge q_f on face f is related to the branch charge on capacitive edge e by $q_e = \sum_{f \in F} B_{f,e} q_f$.

Lemma 6.4 (Face-redundancy of B). *For a connected planar circuit, the orientation matrix satisfies $\langle 1_F | B = \langle 0 |$, equivalently $\sum_{f \in F} B_{f,e} = 0$ for every edge e . Consequently $\text{rank}(B) = |F| - 1$ and $\ker(B^\top) = \text{span}\{|1_F\rangle\}$.*

Proof. Every edge e on S^2 borders exactly two faces, and orienting each face consistently (say counterclockwise) means the two adjacent faces traverse e in opposite directions. The two contributions to $\sum_{f \in F} B_{f,e}$ cancel and all other faces contribute zero, giving $\langle 1_F | B = \langle 0 |$, hence $|1_F\rangle \in \ker(B^\top)$ and $\text{rank}(B) \leq |F| - 1$. The reverse inequality follows from $\text{rank}(B) \geq |F| - 1 = |E| - |V| + 1$, the dimension of the cycle space, which the face boundaries span (remark 6.2). $\square$

Below are the orientation matrices for the circuit examples encountered so far. In each case, $|F| = |E| - |V| + 2$ and the rows satisfy $\sum_f B_{f,e} = 0$ for every edge.

$$\begin{array}{c} \begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} f_1 \\ f_2 \\ f_3 \\ f_{ext} \end{matrix} & \begin{pmatrix} +1 & 0 & 0 & -1 & 0 & +1 \\ 0 & +1 & 0 & 0 & -1 & -1 \\ 0 & 0 & +1 & +1 & +1 & 0 \\ -1 & -1 & -1 & 0 & 0 & 0 \end{pmatrix} \end{matrix} & \begin{matrix} & e_1 & e_2 \\ \begin{matrix} f_1 \\ f_{ext} \end{matrix} & \begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix} \end{matrix} & \begin{matrix} & e_1 & e_2 & e_3 \\ \begin{matrix} f_1 \\ f_{ext} \end{matrix} & \begin{pmatrix} +1 & +1 & +1 \\ -1 & -1 & -1 \end{pmatrix} \end{matrix} \end{array}$$

K_4 Circuit (Figure 1)

The Dualmon (Figure 4) Two Cap. Loop (Figure 5)

Furthermore, the connection between the dual graph co-incidence matrix A^* and B is not hard to intuit. Accordingly, the orientation matrix satisfies a similar fundamental exactness relation.

Lemma 6.5 (Exactness Sequence). *The incidence matrix A and orientation matrix B satisfy*

$$\ker(B) = \text{Im}(A^\top). \quad (6.2)$$

That is, a vector $|w\rangle$ in edge-space is annihilated by B if and only if it is of the form $A^\top|v\rangle$ for some node-space ket $|v\rangle$.

Proof. There are two inclusions to check.

$\text{Im}(A^\top) \subseteq \ker(B)$. We need $BA^\top = 0$. For any node v and face f ,

$$(BA^\top)_{f,v} = \sum_{e \in E} B_{f,e} A_{e,v}^\top = \sum_{e \in E} B_{f,e} A_{v,e}. \quad (6.3)$$

The node v contributes to this sum only through edges incident to it. If v lies on the boundary of f , the boundary enters v along one edge and leaves along another, and these two edges

contribute with opposite signs in A (one has v as head, the other as tail) and the same sign in B (both traversed in the boundary's direction); the terms cancel. If v does not lie on the boundary of f , there is nothing to sum. Either way, $(BA^\top)_{f,v} = 0$.

$\ker(B) \subseteq \text{Im}(A^\top)$. A dimension count closes the gap. We have $\text{rank}(A^\top) = |V| - 1$ (the only relation is the uniform vector $|1\rangle$), and $\text{rank}(B) = |F| - 1$ by [lemma 6.4](#), so

$$\dim \ker(B) = |E| - (|F| - 1) = |V| - 1 = \text{rank}(A^\top) = \dim \text{Im}(A^\top), \quad (6.4)$$

using $|F| = |E| - |V| + 2$. Two subspaces of equal dimension with one contained in the other must be the same. $\square$

6.3. The Edge-Parity Operator and Connection Matrix

We are now closer to defining the eventual matrix that will serve as the symmetric symplectic pairing object. The key idea of the flux-charge symmetric formulation is to distinguish capacitive from inductive edges through an operator that assigns opposite signs to the two types.

Definition 6.6 (Edge-Parity Operator). The edge-parity operator $P \in \mathbb{R}^{|E| \times |E|}$ is diagonal:

$$P_{e,e'} := \begin{cases} +1 & \text{if } e = e' \in E_C, \\ -1 & \text{if } e = e' \in E_L, \\ 0 & \text{if } e \neq e'. \end{cases} \quad (6.5)$$

Equivalently, $P = \text{diag}(+1)_{E_C} + \text{diag}(-1)_{E_L}$. Note that $P^2 = I$, meaning P is an involution, a fact that will become important soon.

Definition 6.7 (Connection Matrix). We now define the symmetric object that has been the topic of much discussion. The connection matrix M of the circuit G is given by

$$M := \frac{1}{2} B P A^\top \in \mathbb{R}^{|F| \times |V|}. \quad (6.6)$$

An explicit separation into capacitive and inductive components finds,

$$M_{f,v} = \frac{1}{2} \sum_{e \in E_C} B_{f,e} A_{e,v}^\top - \frac{1}{2} \sum_{e \in E_L} B_{f,e} A_{e,v}^\top. \quad (6.7)$$

Lemma 6.8 (Equivalent forms of M). *The connection matrix can equivalently be written as*

$$M = \sum_{e \in E_C} B_{(\cdot,e)} A_{(e,\cdot)}^\top = - \sum_{e \in E_L} B_{(\cdot,e)} A_{(e,\cdot)}^\top, \quad (6.8)$$

where $B_{(\cdot,e)}$ denotes the column of B corresponding to edge e and $A_{(e,\cdot)}^\top$ denotes the row of $A^\top$ corresponding to edge e .

Proof. From the exact sequence (6.2), $BA^\top = 0$, which means $\sum_{e \in E} B_{f,e} A_{e,v}^\top = 0$ for all f, v . Therefore $\sum_{e \in E_C} B_{f,e} A_{e,v}^\top = - \sum_{e \in E_L} B_{f,e} A_{e,v}^\top$ and the rest follows from (6.7). $\square$

The connection matrix M maps node fluxes to loop charges, seeing both capacitive and inductive edges through P . Notice that M is symmetric in the sense that swapping $E_C \leftrightarrow E_L$ (i.e., replacing P with $-P$) only changes the sign of M , demonstrating that no type of element is privileged over the other. We may finally define the corresponding circuit action.

6.4. The Symmetric Circuit Action

Definition 6.9 (Circuit Action). Let $G = (V, E, \sigma)$ be a circuit graph with connection matrix M . The circuit action and Lagrangian L are therefore:

$$S[|\phi_V\rangle, |q_F\rangle] = \int d\tau \left[\underbrace{\sum_{f \in F, v \in V} q_f M_{f,v} \dot{\phi}_v - E_{\text{tot}}(|\phi_V\rangle, |q_F\rangle)}_{L[|\phi_V\rangle, |q_F\rangle]} \right], \quad (6.9)$$

where $|\phi_V\rangle = (\phi_v)_{v \in V}$ are the node fluxes and $|q_F\rangle = (q_f)_{f \in F}$ are the face (loop) charges, treated as independent dynamical variables. The total energy is

$$E_{\text{tot}}(|\phi_V\rangle, |q_F\rangle) = \sum_{e \in E_C} \mathcal{E}_{C,e} \left(\sum_{f \in F} B_{f,e} q_f \right) + \sum_{e \in E_L} \mathcal{E}_{L,e} \left((A_L^\top |\phi_V\rangle)_e \right), \quad (6.10)$$

where each $\mathcal{E}_{C,e}$ and $\mathcal{E}_{L,e}$ is a smooth energy function of that element. The $\mathcal{E}$ symbol is used to avoid confusion with the edge sets.

The action (6.9) treats capacitive and inductive elements symmetrically in the same way M does. Swapping the roles of E_C and E_L (taking the dual circuit) amounts to replacing M with $-M$, which simply exchanges the roles of $|\phi_V\rangle$ and $|q_F\rangle$ in the symplectic term. This is the core of flux-charge symmetry. Recall that the action written in terms of the reduced capacitive incidence matrix Ω would change its structure entirely under such a swap.

6.5. The Euler-Lagrange Equations Reproduce the Circuit

Theorem 6.10 (Circuit dynamical systems are Lagrangian). Let $G = (V, E, \sigma)$ be a finite connected circuit graph with smooth constitutive relations. Taking the physical variables to be $(\phi_v, q_f)_{v \in V, f \in F}$ and S to be the circuit action (6.9), the Euler-Lagrange equations

$$\frac{\delta S}{\delta \phi_v} = 0 \quad \forall v \in V, \quad \frac{\delta S}{\delta q_f} = 0 \quad \forall f \in F \quad (6.11)$$

exactly reproduce the circuit dynamical system $\Sigma_e = \Lambda_1 = \Lambda_2 = 0$ (3.17). Therefore, every circuit dynamical system is a Lagrangian dynamical system.

Proof. We compute the two sets of Euler-Lagrange equations.

Variation with respect to ϕ_v . The action (6.9) depends on $\dot{\phi}_v$ through the symplectic term and on ϕ_v through the inductive energy. Computing:

$$\frac{\partial L}{\partial \dot{\phi}_v} = \sum_{f \in F} q_f M_{f,v}, \quad \frac{\partial L}{\partial \phi_v} = - \sum_{e \in E_L} \frac{\partial \mathcal{E}_{L,e}}{\partial \phi_e} \cdot (A_L^\top)_{e,v}.$$

The Euler-Lagrange equation gives, for each $v \in V$:

$$\sum_{f \in F} \dot{q}_f M_{f,v} + \sum_{e \in E_L} \frac{\partial \mathcal{E}_{L,e}}{\partial \phi_e} (A_L^\top)_{e,v} = 0. \quad (6.12)$$

The first sum, using $M_{f,v} = \sum_{e \in E_C} B_{f,e} A_{e,v}^\top$ from lemma 6.8, becomes $\sum_{f,e \in E_C} \dot{q}_f B_{f,e} A_{e,v}^\top =$

$\sum_{e \in E_C} \dot{q}_e A_{e,v}^\top$ where we used $\dot{q}_e = \sum_f B_{f,e} \dot{q}_f$. So (6.12) is

$$\sum_{e \in E_C} \dot{q}_e (A_C)_{v,e} + \sum_{e \in E_L} I_e (A_L)_{v,e} = 0,$$

which is exactly KCL at node v , i.e. $\Lambda_1 = 0$ and the inductive half of $\Sigma_e = 0$.

Variation with respect to q_f . Since $\dot{q}_f$ does not appear in (6.9), the Euler-Lagrange equation reduces to $\partial L / \partial q_f = 0$:

$$\sum_{v \in V} M_{f,v} \dot{\phi}_v = \sum_{e \in E_C} \frac{\partial \mathcal{E}_{C,e}}{\partial q_e} B_{f,e}. \quad (6.13)$$

Using $M_{f,v} = \sum_{e \in E_C} B_{f,e} (A_C^\top)_{e,v}$, the left side is $\sum_{e \in E_C} B_{f,e} (A_C^\top \dot{\phi}_V)_e = \sum_{e \in E_C} B_{f,e} \dot{\phi}_e$, so (6.13) gives

$$\sum_{e \in E_C} B_{f,e} (\dot{\phi}_e - f_e(q_e)) = 0 \quad \forall f \in F.$$

By lemma 6.4, B has rank $|F| - 1$ with kernel spanned by $\langle 1_F |$. The single relation $\sum_f B_{f,e} = 0$ makes the $\langle 1_F |$ component of the system above vacuous, leaving $|F| - 1 = |E| - |V| + 1$ independent equations on the $|E_C|$ unknowns $\dot{\phi}_e - f_e(q_e)$, which are equivalent to $\dot{\phi}_e = f_e(q_e)$ for all $e \in E_C$. This is exactly $\Sigma_e = 0$ on capacitive branches.

The two sets of equations together are precisely $\Sigma_e = \Lambda_1 = \Lambda_2 = 0$. $\square$

7. THE SYMPLECTIC OBSTRUCTION: NULL VECTORS OF M

By now, we have established through theorem 6.10, that every circuit dynamical system arises from the action (6.9). The next step is to pass from this Lagrangian to a Hamiltonian formulation. The path to do so requires identifying canonical conjugate pairs. As we have hinted at, and is perhaps evident by now, the natural candidates from (6.9) are the loop/face charges q_f and node fluxes ϕ_v , coupled through M . If M were a square invertible matrix, the pairing $\{q_f, \phi_v\} = M_{fv}^{-1}$ would follow immediately, and we would be done. However, M is generally neither square nor invertible (a complication that will slightly alter our choice of coordinates), and we must understand its null structure before proceeding.

7.1. The Exactness Lemma

The structure of the null space of M is controlled by a general algebraic exactness lemma, the importance of which cannot be overstated. Recall from definition 6.6 that P is an involution: $P^2 = I$. This means the only eigenvalues of P are ± 1 , and any vector in edge-space decomposes uniquely into its $+1$ and -1 eigenspaces (corresponding to capacitive and inductive edges respectively).

Lemma 7.1 (Exactness). *Let $A : X \rightarrow Y$ and $B : Y \rightarrow Z$ be linear maps satisfying $\ker(B) = \text{Im}(A)$, and let $P : Y \rightarrow Y$ be an involution. If $|v\rangle \in X$ satisfies $BPA|v\rangle = |0\rangle$, then there exist $|u_+\rangle, |u_-\rangle \in X$ such that $|v\rangle = |u_+\rangle + |u_-\rangle$ and $A|u_\pm\rangle$ is a ± 1 eigenvector of P . Moreover, $A|u_+\rangle$ and $A|u_-\rangle$ are separately in $\ker(B)$.*

Proof. Since $P^2 = I$, every $|w\rangle \in Y$ decomposes as

$$|w\rangle = |w\rangle_+ + |w\rangle_-, \quad |w\rangle_\pm := \frac{1}{2}(|w\rangle \pm P|w\rangle),$$

with $P|w\rangle_{\pm} = \pm|w\rangle_{\pm}$. Apply this to $A|v\rangle$:

$$A|v\rangle = (A|v\rangle)_+ + (A|v\rangle)_-.$$

From $BPA|v\rangle = |0\rangle$:

$$|0\rangle = B(P(A|v\rangle)_+ + P(A|v\rangle)_-) = B((A|v\rangle)_+ - (A|v\rangle)_-),$$

so $(A|v\rangle)_+ - (A|v\rangle)_- \in \ker(B) = \text{Im}(A)$. Thus there exists $|w\rangle \in X$ with $A|w\rangle = (A|v\rangle)_+ - (A|v\rangle)_-$. Set

$$|u_+\rangle := \frac{1}{2}(|v\rangle + |w\rangle), \quad |u_-\rangle := \frac{1}{2}(|v\rangle - |w\rangle).$$

Then $|u_+\rangle + |u_-\rangle = |v\rangle$, and $A|u_+\rangle = (A|v\rangle)_+$ is a $+1$ eigenvector of P , $A|u_-\rangle = (A|v\rangle)_-$ is a -1 eigenvector of P . Finally, $B(A|v\rangle)_+ = B(A|v\rangle)_- = |0\rangle$ follows from the two equations above, so both $A|u_{\pm}\rangle \in \ker(B) = \text{Im}(A)$. $\square$

We will eventually see that this lemma will be our main tool for characterizing the null space of M , but it also offers some preliminary intuition for why that null space takes on such a structured form. In general, the null space of a matrix can be computationally difficult to describe, so embarking on this null space characterization a priori must be justified. The exactness lemma does exactly this: any vector killed by a composition BPA , where P is an involution and B, A are exact as in [lemma 6.5](#), must decompose into pieces that are separately killed by B , and each piece is composed of only one type of edge (capacitive or inductive) through P . Note that our connection matrix M matches this form with $A \rightarrow A^\top$. This is the mechanism that will identify the null vectors of M with homogeneous graph structures.

7.2. Homogeneous Cuts and Loops

The asymmetric generalization of the inductively shunted island and capacitive loop (see [Section A.5](#)) is the notion of a *homogeneous* cut or loop, meaning made entirely of one type of element.

Definition 7.2 (Homogeneous Cut). A cut of G is a partition (V_1, V_2) of the node set V , with the cut set $\mathfrak{C}(V_1, V_2) := \{e \in E : e \text{ has one endpoint in each of } V_1, V_2\}$. A cut is homogeneous if $\mathfrak{C}(V_1, V_2) \subseteq E_C$ (all capacitive branches) or $\mathfrak{C}(V_1, V_2) \subseteq E_L$ (all inductive branches). Denote Γ_C for the set of capacitively homogeneous cuts and Γ_L for the set of inductively homogeneous cuts. We adopt the convention that the empty cut $(V, \emptyset)$ is vacuously both a capacitive and an inductive cut (a different convention than [\[6\]](#), [\[7\]](#)).

Definition 7.3 (Homogeneous Loop). A *loop* of G , in the sense used here, is any subset $\gamma \subseteq E$ of edges that forms a closed walk—equivalently, any element of the cycle space $\ker(A)$. A loop is homogeneous if $\gamma \subseteq E_C$ (capacitively homogeneous) or $\gamma \subseteq E_L$ (inductively homogeneous). Write Δ_C for the set of capacitively homogeneous loops and Δ_L for the inductively homogeneous ones. See [remark 3.16](#) to see the connection between faces and loops.

Identifying homogeneous objects on a circuit graph is important, as we will see they correspond to null directions. For complicated circuits, enumerating them by hand can be difficult. We will eventually find that, through another graphical object, the spanning tree, identifying them manually is not necessary. As an example, we examine a circuit built on the triangular prism graph [Figure 6](#).

Remark 7.4 (Homogeneous loops vs. face boundaries). A homogeneous loop need not be a face boundary. It may be any closed walk supported entirely on E_C (or E_L): for instance, the signed sum of two face boundaries whose shared non-homogeneous edges cancel can be a homogeneous loop even when neither face is itself a homogeneous boundary. Faces serve only as the (redundant) coordinate system in which loop charges live; the homogeneous content is intrinsic to G and its σ -coloring.

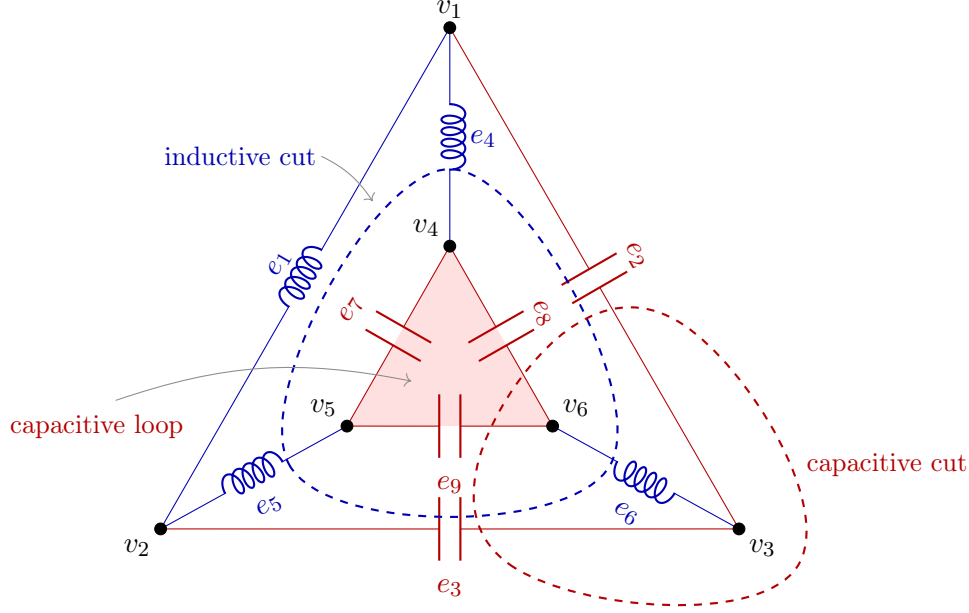


Figure 6: This circuit illustrates three of the four types of homogeneous objects. The inner triangle $\{e_7, e_8, e_9\}$ is a capacitively homogeneous loop (shaded). The three spokes $\{e_4, e_5, e_6\}$ form an inductively homogeneous cut. The four capacitors $\{e_2, e_3, e_8, e_9\}$ form a capacitive cut.

7.2.1. Physical Meaning

An inductively homogeneous cut corresponds to a group of nodes that are connected to the rest of the circuit exclusively through inductors (definition A.1). Shifting all node fluxes within this group by the same constant changes no capacitive branch voltage, which measures the relative difference. The inductive contribution cancels in the symplectic term $\sum_{f,v} q_f M_{f,v} \dot{\phi}_v$ because M sees inductive edges with opposite sign. By symmetry, the same can be said for capacitively homogeneous cuts.

A homogeneous loop is a closed path made entirely of one type of element. For a purely capacitive loop $\gamma \in \Delta_C$, the loop coordinate $|r_\gamma\rangle$ satisfies

$$\langle r_\gamma | M = \langle r_\gamma | B P_C A^\top = \langle \gamma | A^\top = 0, \quad (7.1)$$

since the loop enters and leaves each node with opposite signs in A . The same argument shows that a purely inductive loop is non-dynamical. Physically speaking, the circulating charge produces no charge accumulation at any given node, and is thus “invisible.”

Note how these definitions generalize the notions used in the analysis of Ω in Section A.5. What was previously called an “inductively shunted island” is a connected component of the capacitive subgraph alone, visible only to Ω . A homogeneous cut, by contrast, is a partition of the *full* graph whose cut set is purely capacitive or purely inductive. Similarly, a homogeneous loop is a closed walk in the full graph supported entirely on one type of edge, not just those capacitive ones seen by the reduced capacitive matrix Ω . With these mechanisms at hand and the exactness lemma established, we can now fully characterize the degeneracy of M graph-theoretically.

7.3. Characterizing the Null Space of M

Theorem 7.5 (Null vectors of M). *Let $G = (V, E, \sigma)$ be a connected planar circuit graph with connection matrix $M = \frac{1}{2}BPA^\top$.*

1. *Right null space: A ket $|\phi_V\rangle \in \mathbb{R}^{|V|}$ satisfies $M|\phi_V\rangle = |0\rangle$ if and only if $|\phi_V\rangle$ decomposes into constants on each connected component of the capacitive subgraph (V, E_C) and inductive subgraph (V, E_L) . Equivalently,*

$$|\phi_V\rangle = \sum_{\gamma \in \Gamma_C \cup \Gamma_L} \beta_\gamma |1_\gamma\rangle, \quad (7.2)$$

where each $\beta_\gamma \in \mathbb{R}$, and for each homogeneous cut $\gamma = (V_1, V_2)$ the indicator ket $|1_\gamma\rangle \in \mathbb{R}^{|V|}$ has components

$$(1_\gamma)_v = \begin{cases} 1 & v \in V_1, \\ 0 & v \in V_2. \end{cases} \quad (7.3)$$

2. *Left null space: A ket $|r\rangle \in \mathbb{R}^{|F|}$ satisfies $\langle r|M = \langle 0|$ if and only if $B^\top|r\rangle$ is either zero or a homogeneous cycle in $\Delta_C \cup \Delta_L$. Equivalently, $|r\rangle$ admits the decomposition*

$$|r\rangle = \alpha_0 |1_F\rangle + \sum_{\gamma \in \Delta_C \cup \Delta_L} \alpha_\gamma |r_\gamma\rangle, \quad (7.4)$$

where $\alpha_0, \alpha_\gamma \in \mathbb{R}$, the term $\alpha_0 |1_F\rangle$ is the face-redundancy direction in $\ker(B^\top)$ (lemma 6.4), and each $|r_\gamma\rangle \in \mathbb{R}^{|F|}$ satisfies $B^\top|r_\gamma\rangle = |\gamma\rangle$ (unique modulo $|1_F\rangle$).

3. *Rank Counting: The rank of M is*

$$\text{rank}(M) = |F| - |\Delta_L| - |\Delta_C| - 1 = |V| - |\Gamma_L| - |\Gamma_C| + 1, \quad (7.5)$$

where $|\Delta_C|, |\Delta_L|$ denote the dimensions of the capacitive and inductive cycle spaces, and $|\Gamma_C|, |\Gamma_L|$ denote the dimensions of the subspaces of node-space spanned by capacitively and inductively homogeneous cut indicators.

Proof. We prove each part in turn. Throughout, we use lemma 6.5 in the form $\ker(B) = \text{Im}(A^\top)$, and its transpose:

$$\ker(A) = \text{Im}(B^\top). \quad (7.6)$$

The transpose claim follows from $AB^\top = (BA^\top)^\top = 0$ and analogous dimension counting.

Part 1.

($\Leftarrow$) Suppose $|\phi_V\rangle$ is constant on each connected component of (V, E_C) and on each connected component of (V, E_L) . For any capacitive edge $e = (u \rightarrow v) \in E_C$, the endpoints u, v lie in the same capacitive component, so $(A^\top|\phi_V\rangle)_e = \phi_v - \phi_u = 0$. Similarly on E_L , $(A^\top|\phi_V\rangle)_e = 0$ for every $e \in E_L$ as well. Hence $A^\top|\phi_V\rangle = |0\rangle$, and immediately, $M|\phi_V\rangle = \frac{1}{2}BPA^\top|\phi_V\rangle = |0\rangle$.

The explicit decomposition is not difficult to see, but it runs the risk of heavy notation. Let $\{U_1^C, \dots, U_{|\Gamma_L|}^C\}$ be the connected components of (V, E_C) and $\{U_1^L, \dots, U_{|\Gamma_C|}^L\}$ those of (V, E_L) . The cut $\gamma_i^C = (U_i^C, V \setminus U_i^C)$ has cut-set contained in E_L , so $\gamma_i^C \in \Gamma_L$; analogously $\gamma_j^L \in \Gamma_C$. The indicators $\{|1_{\gamma_i^C}\rangle\} \cup \{|1_{\gamma_j^L}\rangle\}$ span exactly the vectors constant on both subgraphs, so every such $|\phi_V\rangle$ admits the representation (7.2).

($\Rightarrow$) Conversely, suppose $M|\phi_V\rangle = |0\rangle$, meaning $BPA^\top|\phi_V\rangle = |0\rangle$. Apply [lemma 7.1](#) with the roles $A \mapsto A^\top$ and P as our edge-parity operator, which is an involution. Then there exist $|u_+\rangle, |u_-\rangle$ with

$$|\phi_V\rangle = |u_+\rangle + |u_-\rangle, \quad PA^\top|u_\pm\rangle = \pm A^\top|u_\pm\rangle, \quad BA^\top|u_\pm\rangle = |0\rangle. \quad (7.7)$$

The eigenvalue condition $PA^\top|u_+\rangle = +A^\top|u_+\rangle$ means $A^\top|u_+\rangle$ is supported on E_C (where $P = +I$). This tells us that $(A^\top|u_+\rangle)_e = 0$ for every $e \in E_L$. Thus, we have that the ket $|u_+\rangle$ is constant on each connected component of (V, E_L) .

Likewise, $A^\top|u_-\rangle$ is supported on E_L , so $A^\top|u_-\rangle = |0\rangle$ and $|u_-\rangle$ is constant on each capacitive component. Writing each in its component-indicator basis,

$$|u_+\rangle = \sum_j \beta_{\gamma_j^L} |1_{\gamma_j^L}\rangle, \quad |u_-\rangle = \sum_i \beta_{\gamma_i^C} |1_{\gamma_i^C}\rangle, \quad (7.8)$$

which sums to the form [\(7.2\)](#).

Part 2.

($\Leftarrow$) The two summands in [\(7.4\)](#) are handled separately. For $\alpha_0|1_F\rangle$: $\langle 1_F|M = \frac{1}{2}\langle 1_F|BPA^\top = \frac{1}{2}\langle 0|PA^\top = 0$ by [lemma 6.4](#), so $|1_F\rangle \in \ker(M^\top)$. For the cycle terms: suppose $B^\top|r_\gamma\rangle = |\gamma\rangle$ with $|\gamma\rangle$ a homogeneous cycle, i.e. $A|\gamma\rangle = 0$ and $P|\gamma\rangle = \pm|\gamma\rangle$. Then

$$M^\top|r_\gamma\rangle = \frac{1}{2}APB^\top|r_\gamma\rangle = \frac{1}{2}AP|\gamma\rangle = \pm\frac{1}{2}A|\gamma\rangle = |0\rangle. \quad (7.9)$$

By linearity, every $|r\rangle$ of the form [\(7.4\)](#) satisfies $\langle r|M = \langle 0|$.

($\Rightarrow$) Suppose $\langle r|M = \langle 0|$, equivalently $APB^\top|r\rangle = |0\rangle$. The dual exactness [\(7.6\)](#) provides the hypothesis needed to apply exactness [lemma 7.1](#) with the roles $A \mapsto B^\top$ and $B \mapsto A$. The lemma gives a decomposition $|r\rangle = |s_+\rangle + |s_-\rangle$ with $PB^\top|s_\pm\rangle = \pm B^\top|s_\pm\rangle$ and $AB^\top|s_\pm\rangle = |0\rangle$.

Following the same reasoning as Part 1, $B^\top|s_+\rangle$ is a cycle supported on E_C (a capacitive homogeneous cycle), and $B^\top|s_-\rangle$ an inductive one. By [lemma 6.4](#), $B^\top$ has kernel spanned by $|1_F\rangle$, so each cycle of this form determines $|s_\pm\rangle$ uniquely modulo $\text{span}\{|1_F\rangle\}$. Choosing representatives and absorbing the residual $|1_F\rangle$ component into a coefficient α_0 , the expansion of $|s_+\rangle$ and $|s_-\rangle$ in bases of Δ_C and Δ_L assembles to [\(7.4\)](#).

Part 3. We count $\dim \ker(M)$ in two ways.

First, from Part 1, $\ker(M) = \ker(A_C^\top) + \ker(A_L^\top)$. By the component characterization of each kernel, $\dim \ker(A_C^\top) = |\Gamma_L|$ and $\dim \ker(A_L^\top) = |\Gamma_C|$. Their intersection consists of vectors constant on every capacitive and inductive component, which by connectivity of G is the entire graph—the empty cut by our convention ([definition 7.2](#)). Therefore $\ker(A_C^\top) \cap \ker(A_L^\top) = \text{span}\{|1_V\rangle\}$ is one-dimensional. Inclusion-exclusion gives:

$$\dim \ker(M) = |\Gamma_L| + |\Gamma_C| - 1, \quad (7.10)$$

and by rank-nullity,

$$\text{rank}(M) = |V| - |\Gamma_L| - |\Gamma_C| + 1. \quad (7.11)$$

Second, from Part 2, $\ker(M^\top)$ is spanned by the face-redundancy direction $|1_F\rangle$ together with the homogeneous cycle representatives. The cycle representatives $\{|r_\gamma\rangle : \gamma \in \Delta_C \cup \Delta_L\}$ are independent modulo $|1_F\rangle$ (since their images $|\gamma\rangle$ under $B^\top$ are independent in $\ker(A)$, and the

homogeneous cycle sets are disjoint as the empty cycle is not counted), so

$$\dim \ker(M^\top) = |\Delta_L| + |\Delta_C| + 1. \quad (7.12)$$

Applying rank-nullity to $M^\top$ gives

$$\text{rank}(M) = |F| - |\Delta_L| - |\Delta_C| - 1. \quad (7.13)$$

Combining (7.11) and (7.13) completes the proof. $\square$

Corollary 7.6 (Degree of freedom count). *For a connected planar circuit graph,*

$$|F| - |\Delta_L| - |\Delta_C| = |V| - |\Gamma_L| - |\Gamma_C| + 2. \quad (7.14)$$

This common value, equal to $\text{rank}(M) + 1$, counts the physical degrees of freedom in the circuit plus one face-redundancy direction.

Remark 7.7 (Back to Euler). The constant $+2$ on the right of (7.14) is exactly Euler's formula on S^2 : $|V| - |E| + |F| = 2$. The corollary therefore reads as the Euler characteristic decomposed across the symplectic structure: the $+2$ accounts for the uniform node direction $|1_V\rangle \in \ker(A_C^\top) \cap \ker(A_L^\top)$ and the uniform face direction $|1_F\rangle \in \ker(B^\top)$.

Looking back on the linear circuit of Section 4.2, the Legendre transform requires only that $C = A_C C_b A_C^\top$ be non-degenerate on $\mathbb{R}^{|V|}/\ker C$. By letting $k = |\Gamma_L|$ denote the number of connected components of the capacitive subgraph (V, E_C) , we have $\dim \ker C = k$ and the reduced phase space has dimension $|V| - k$ (theorem 4.3). The linear approach eliminates branch charges algebraically via $q_e = C_e \dot{\phi}_e$, which is why they do not appear as independent variables.

It is worth comparing this dimension count to the rank of M from theorem 7.5, as it generalizes to all smooth constitutive relations. When the inductive subgraph is connected ($|\Gamma_C| = 1$), theorem 7.5 gives $\text{rank}(M) = |V| - k - 1 + 1 = |V| - k$, a match. When (V, E_L) is disconnected, $\text{rank}(M) < |V| - k$. M sees the additional non-dynamical directions on the inductive side that it is able to quotient out. By eliminating branch charges from the symplectic term, their contributions appear solely through the energy functional and the linear construction can see only the capacitive null space $\ker C$.

Both descriptions yield equivalent physics, but the M -based count reports the dimension of only the dynamically active phase space directly, whereas the linear approach does not, similar to how the Ω formulation also requires additional bookkeeping.

Example 7.8 (Two capacitors in series: the capacitively shunted island). We verify theorem 7.5 on the two-capacitor circuit introduced in Section 6.1. This graph has $|V| = 3$, $|E| = 3$, $|F| = 2$.

$$\begin{array}{ccc} \begin{array}{c} f_1 \\ f_{ext} \end{array} \begin{array}{c} e_1 \quad e_2 \quad e_3 \\ \left(\begin{array}{ccc} +1 & +1 & +1 \\ -1 & -1 & -1 \end{array} \right) \end{array} & \begin{array}{c} e_1 \quad e_2 \quad e_3 \\ \left(\begin{array}{ccc} +1 & 0 & 0 \\ 0 & +1 & 0 \\ 0 & 0 & -1 \end{array} \right) \end{array} & \begin{array}{c} v_1 \quad v_2 \quad v_3 \\ \left(\begin{array}{ccc} -1 & +1 & 0 \\ 0 & -1 & +1 \\ +1 & 0 & -1 \end{array} \right) \end{array} \\ B & P & A^\top \end{array}$$

$$\text{The connection matrix is therefore } M = \frac{1}{2} B P A^\top = \begin{array}{c} f_1 \\ f_{ext} \end{array} \begin{array}{c} v_1 \quad v_2 \quad v_3 \\ \left(\begin{array}{ccc} -1 & 0 & +1 \\ +1 & 0 & -1 \end{array} \right) \end{array}$$

Homogeneous cuts. The capacitive subgraph is fully connected, so $\ker(A_C^\top) = \text{span}\{(1, 1, 1)\}$, and $|\Gamma_L| = 1$ (the empty cut). $|\Gamma_C| = 2$: the trivial cut, and the cut isolating the inductive island indicator $(1, 0, 1)$ (or equivalently $(0, 1, 0)$).

Homogeneous loops. The capacitive subgraph is a tree (no cycles), so $|\Delta_C| = 0$. The inductive subgraph has a single edge and no cycles, so $|\Delta_L| = 0$.

Rank. The theorem gives $\text{rank}(M) = |V| - |\Gamma_L| - |\Gamma_C| + 1 = 3 - 1 - 2 + 1 = 1$, equivalently $|F| - |\Delta_L| - |\Delta_C| - 1 = 2 - 0 - 0 - 1 = 1$. Direct inspection confirms the two rows of M are negatives of each other ($|1_F\rangle$ is the face redundancy in $\ker(M^\top)$), so $\text{rank}(M) = 1$.

Physical interpretation. The $\text{rank}(M) = 1$ tells us there exists precisely one canonical pair: the charge Q flowing in the loop and the flux $\Phi = \phi_3 - \phi_1$ across the JJ. The two face charges $q_{f_1}, q_{f_{ext}}$ are bookkeeping coordinates with one redundancy ($q_{f_1} + q_{f_{ext}}$ is gauge), the two capacitor charges are related by the Noether constraint (Figure 5), and these collapse into a single dynamical loop charge in the spanning-tree reduction of Section 8.

8. CANONICAL VARIABLES: THE SPANNING TREE CONSTRUCTION

By theorem 7.5, the connection matrix M has non-trivial null spaces on both sides, corresponding to non-dynamical degrees of freedom in the action (6.9). The Hamiltonian description proof of Section 9 uses only the fact that these null spaces have a known dimension and a clean characterization in terms of homogeneous structures; it does not require a derivation of the canonical pairs. From the point of view of dynamical system characterization, this section is therefore optional. What it does provide is an explicit graph-theoretic approach to identifying the canonical pairs by hand—rather than treating the coordinate choice as physics lore. The aim is to construct, given a circuit graph $G = (V, E, \sigma)$, an explicit set of pairs $(\Phi_t, Q_t)_{t \in T}$ that diagonalize the symplectic term, demonstrating that conjugate pairs are intrinsic to the graph topology.

The construction uses a spanning tree of the *capacitive* subgraph (V, E_C) . This will look at first like a regression: we have spent the previous sections insisting that capacitors and inductors must be treated symmetrically, and now are about to single out the capacitive edges. The next subsection explains why this is allowed. In short, the same matrix M admits two equivalent presentations, one capacitively biased and one inductively biased, and the spanning-tree machinery runs on either side and produces canonically equivalent coordinates. In other words, this asymmetric treatment is encompassed by a symmetric model of M , making the choice of a capacitive focus arbitrary.

8.1. An Asymmetric Proof Under a Symmetric Model

The connection matrix as we have defined it in definition 6.7 is, manifestly symmetric in E_C and E_L . The parity operator P assigns $+1$ to capacitive edges and -1 to inductive ones, and swapping the two sends $M \mapsto -M$. The trick is that M can also be written using only one type of edge. Decompose the parity operator using the projections P_C, P_L onto E_C and E_L defined as follows:

$$(P_C)_{ee'} = \mathbf{1}[e \in E_C] \delta_{ee'}, \quad (P_L)_{ee'} = \mathbf{1}[e \in E_L] \delta_{ee'}. \quad (8.1)$$

Since $P_C + P_L = I$ the identity,

$$P = P_C - P_L = P_C - (I - P_C) = 2P_C - I = I - 2P_L. \quad (8.2)$$

Substituting into M and using the exactness identity $BA^\top = 0$ from lemma 6.5,

$$M = \frac{1}{2} B(2P_C - I)A^\top = B P_C A^\top - \frac{1}{2} \underbrace{B A^\top}_{=0} = B P_C A^\top = -B P_L A^\top. \quad (8.3)$$

Notice that lemma 6.8 is exactly this statement. The capacitive presentation drops the inductive edges from the sum; the inductive presentation drops the capacitive ones. Each presentation looks

asymmetric, but fits under this symmetric model.

This is what permits the construction below. We will proceed by fixing a spanning tree on the capacitive subgraph and use only the capacitive presentation $M = BP_C A^\top$ to identify canonical pairs. The mirror construction—spanning tree on the inductive subgraph, using $M = -BP_L A^\top$ —runs through identical steps and produces canonical pairs related to ours by flux-charge duality. Both proofs extract the rank of the same M from two different “slicings” of the same edge space; the asymmetry lives in the presentation, not in the underlying object.

A second reason for working capacitively is mechanical. Restricting to one subgraph makes all manipulations simpler, and is the main focus of [6]. Namely, the symplectic term can be easily rewritten in (8.8) without needing to consider the parity operator or factor of $\frac{1}{2}$. Mixing element types is possible through the notion of a combined spanning forest of capacitive spanning trees and inductive spanning trees, but requires unnecessarily complex manipulations.

8.2. The Spanning Tree of the Capacitive Subgraph

The capacitive subgraph (V, E_C) is the network connecting nodes by capacitive branches alone. A spanning tree of this network is, intuitively, a minimal backbone that connects every capacitively-reachable node without ever closing a loop.

Definition 8.1 (Spanning Tree of the Capacitive Subgraph). A spanning tree of the capacitive subgraph is a subset $T \subseteq E_C$ of capacitive branches such that:

1. *Acyclic*: T contains no cycles. Equivalently, between any two nodes lying in the same connected component of T , there is a unique path using only branches of T .
2. *Spanning*: Every node adjacent to some branch in E_C is adjacent to some branch in T .

Strictly speaking, when (V, E_C) is disconnected, T is a spanning *forest* (hence no connectivity requirement). To the disappointment of graph theorists, we will use the term “tree” in both cases.

A spanning tree of (V, E_C) always exists, and elementary graph theory (Section A.3) shows

$$|T| = |V| - |\Gamma_L|, \quad (8.4)$$

where Γ_L is the set of inductively homogeneous cuts. Note that $A_C^\top = \Omega$ (A.13), but we will refrain from using the Ω notation to emphasize that the construction below is not asymmetrically biased. We illustrate the construction of a capacitive spanning tree on a few running examples in Figure 7.

Fix some such T (the results are independent of the chosen T , theorem 8.3). The reason it solves both null-space problems becomes visible by examining what it does to fluxes and charges separately.

1. *Fluxes*. For each branch $t = (u \rightarrow v) \in T$, define the tree branch flux

$$\Phi_t := \phi_v - \phi_u = (A_C^\top |\phi_V\rangle)_t. \quad (8.5)$$

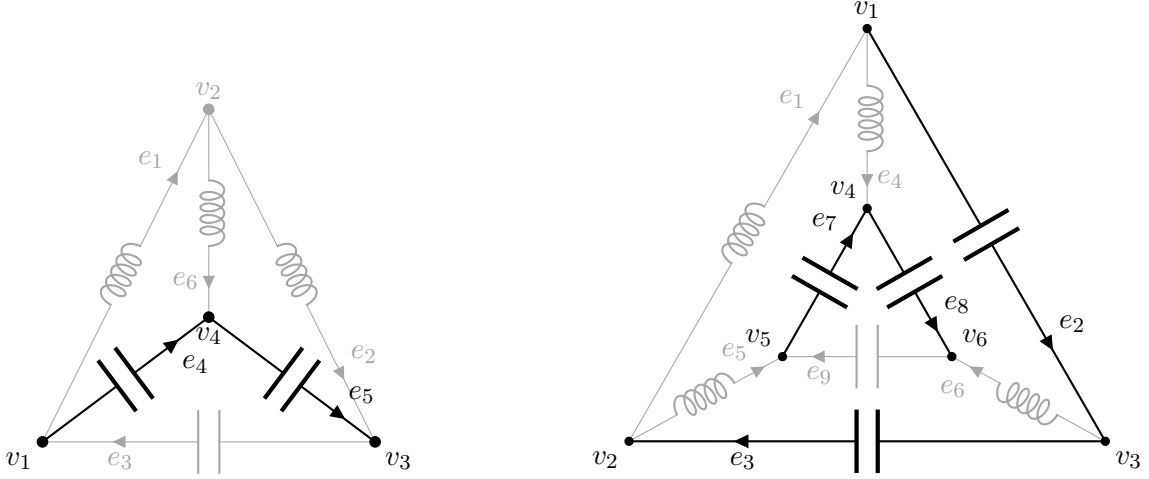
If $|\phi_V\rangle \in \ker(A_C^\top)$, that is, $|\phi_V\rangle$ admits a constant value on each inductively homogeneous cut (theorem A.3), then every Φ_t vanishes. Conversely, if all Φ_t vanish on the tree, then $|\phi_V\rangle$ is constant on each connected component of (V, E_C) , so $|\phi_V\rangle \in \ker(A_C^\top)$. The map $|\phi_V\rangle \mapsto (\Phi_t)_{t \in T}$ therefore has kernel exactly equal to $\ker(A_C^\top)$, and the $|T|$ tree fluxes are linearly independent modulo $\ker(A_C^\top)$.

2. *Charges*. Each capacitive co-tree branch $e \in E_C \setminus T$, when added to T , closes a unique fundamental cycle. The total signed branch-charge contribution to tree branch t from all

co-tree capacitive branches is given by

$$\sum_{e \in E_C \setminus T} \sigma_{et} q_e, \quad \sigma_{et} \in \{-1, 0, 1\}, \quad (8.6)$$

where $\sigma_{et} = +1$ if t is traversed in its positive direction along the fundamental cycle of e , -1 if reversed, and 0 if t does not lie on that cycle. These cycle-induced contributions are *not* independent degrees of freedom but linear combinations constrained by Kirchhoff's laws, and this redundancy is what the new charge coordinate Q_t defined below will absorb.



(a) The K_4 circuit of Figure 1 with capacitive spanning tree $T = \{e_4, e_5\}$.

(b) The triangular prism circuit of Figure 6 with spanning tree (forest) $T = \{e_2, e_3, e_7, e_8\}$.

Figure 7: Two spanning tree choices (highlighted with thick lines) on circuits introduced earlier. In each case, each co-tree capacitive branch closes a unique fundamental cycle with T .

8.3. The Canonical Coordinates

The tree fluxes $\{\Phi_t\}_{t \in T}$ are the canonical flux coordinates, independent by the argument above. The branch charges $\{q_t\}_{t \in T}$, however, are *not* the right charge coordinates: Kirchhoff's current law and the cycle constraints relate them to the co-tree charges, so they are not independent. We introduce a new variable Q_t that bundles each tree branch charge together with its co-tree contributions.

Definition 8.2 (Conjugate Charge). Given a spanning tree T as in definition 8.1, the conjugate charge of a tree branch $t \in T$ is

$$Q_t := q_t + \sum_{e \in E_C \setminus T} \sigma_{et} q_e, \quad (8.7)$$

with σ_{et} the orientation indicators of (8.6).

8.4. The Spanning Tree Theorem

Theorem 8.3 (Spanning tree construction [6]). Let $T \subseteq E_C$ be a spanning tree of the capacitive subgraph. Define Φ_t and Q_t as in (8.5, 8.7). Then:

1. The capacitive symplectic term simplifies to

$$\sum_{e \in E_C, v \in V} q_e (A_C^\top)_{e,v} \dot{\phi}_v = \sum_{t \in T} Q_t \dot{\Phi}_t. \quad (8.8)$$

The left side involves $|E_C|$ branch charges and $|V|$ node fluxes through a non-invertible matrix; the right side has exactly $|T|$ canonical pairs with no redundancy.

2. The pairs are canonically conjugate:

$$\{\Phi_t, Q_{t'}\} = \delta_{tt'}, \quad \{\Phi_t, \Phi_{t'}\} = 0, \quad \{Q_t, Q_{t'}\} = 0. \quad (8.9)$$

3. The choice of spanning tree does not matter: any two spanning trees are canonically equivalent.

Proof. We proceed part by part.

Part 1 (symplectic simplification). In the same way we decomposed charge, we can apply KVL to a fundamental cycle of each co-tree branch, and its fluxes decompose along the tree as

$$\Phi_e = \sum_{t \in T} \sigma_{et} \Phi_t \quad (8.10)$$

We can then analyze the symplectic term by splitting the sum over $E_C = T \cup (E_C \setminus T)$ and substituting (8.10):

$$\sum_{e \in E_C, v} q_e (A_C^\top)_{e,v} \dot{\phi}_v = \sum_{t \in T} q_t \underbrace{\sum_v (A_C^\top)_{t,v} \dot{\phi}_v}_{=\dot{\Phi}_t} + \sum_{e \in E_C \setminus T} q_e \underbrace{\sum_v (A_C^\top)_{e,v} \dot{\phi}_v}_{=\dot{\Phi}_e} \quad (8.11)$$

$$= \sum_{t \in T} q_t \dot{\Phi}_t + \sum_{e \notin T} q_e \sum_{t \in T} \sigma_{et} \dot{\Phi}_t \quad (8.12)$$

$$= \sum_{t \in T} \left(q_t + \sum_{e \notin T} \sigma_{et} q_e \right) \dot{\Phi}_t \quad (8.13)$$

$$= \sum_{t \in T} Q_t \dot{\Phi}_t. \quad (8.14)$$

The $|T|$ pairs (Φ_t, Q_t) are linearly independent by the previous subsections, so (8.14) is a non-degenerate diagonal pairing.

Part 2 (canonical conjugacy). With Part 1 in hand, the circuit action (6.9) restricted to the capacitive presentation reads $S = \int d\tau [\sum_t Q_t \dot{\Phi}_t - H]$. The bracket $\{\Phi_t, Q_{t'}\} = \delta_{tt'}$ then follows directly from the Lagrangian structure exactly as in Section 4.1. The full derivation is given in Section A.4.

Part 3 (independence of spanning tree). We show that two spanning trees produce coordinates related by a canonical transformation (physically equivalent). The argument has three stages: (i) construct the change-of-basis matrix R relating the two sets of tree fluxes; (ii) determine the corresponding charge transformation; (iii) verify that the symplectic term is preserved.

Single-edge swap. Consider first the case where two capacitive spanning trees T and T' differ at a single branch: $T = A \cup \{t_0\}$ and $T' = A \cup \{t_1\}$ with $A \subseteq E_C$ and $t_0, t_1 \notin A$. Adding t_1 to T creates a unique fundamental cycle l' , and t_0 must lie on this cycle ($\sigma_{t_1 t_0}(T) \neq 0$).

The change-of-basis matrix R . Branches in $A = T \cap T'$ are shared, so $\Phi_t = \Phi'_t$ for $t \in A$. The only nontrivial relation involves t_0 and t_1 . Applying (8.10) to the cycle l' ,

$$\Phi_{t_1} = \sum_{t \in T} \sigma_{t_1 t}(T) \Phi_t, \quad (8.15)$$

Solving for Φ_{t_0} , and noticing $\sigma \in \{\pm 1\}$,

$$\sigma_{t_1 t_0}(T) \Phi_{t_0} = \Phi_{t_1} - \sum_{t \neq t_0, t \in T} \sigma_{t_1 t}(T) \Phi_t \implies \Phi_{t_0} = \sigma_{t_1 t_0}(T) \Phi_{t_1} - \sigma_{t_1 t_0}(T) \sum_{t \neq t_0, t \in T} \sigma_{t_1 t}(T) \Phi_t \quad (8.16)$$

Therefore, there exists some square integer matrix $R \in \mathbb{R}^{(|V|-1) \times (|V|-1)}$ such that

$$\Phi_t = \sum_{t' \in T'} R_{tt'} \Phi_{t'} \quad \forall t \in T. \quad (8.17)$$

In block form, with T ordered as $A \cup \{t_0\}$ and T' as $A \cup \{t_1\}$, one can check that

$$R = \begin{pmatrix} I_{|A|} & 0 \\ 0 & \sigma_{t_1 t_0}(T) \end{pmatrix} \begin{pmatrix} I_{|A|} & 0 \\ -\sigma_{t_1 t}(T) & 1 \end{pmatrix} =: D(I + N), \quad (8.18)$$

where D is diagonal with ± 1 entries (so $D^2 = I$) and N is strictly lower triangular with $N^2 = 0$. The factorization gives $R^{-1} = (I - N)D$. This means flux coordinates transform by an invertible integer matrix for capacitive spanning trees that differ by one branch.

The charge transformation. Substituting $\dot{\Phi}_t = \sum_{t'} R_{tt'} \dot{\Phi}'_{t'}$ into the symplectic term,

$$\sum_{t \in T} Q_t \dot{\Phi}_t = \sum_{t' \in T'} \left(\sum_{t \in T} Q_t R_{tt'} \right) \dot{\Phi}'_{t'}. \quad (8.19)$$

Then, the canonical transformation immediately follows from the following assignment:

$$Q'_{t'} = \sum_{t \in T} Q_t R_{tt'}, \quad \text{equivalently} \quad Q' = Q R, \quad (8.20)$$

Together, $Q \mapsto QR$ and $\Phi \mapsto R^{-1}\Phi$ preserve the symplectic term.

General spanning trees. For two spanning trees differing by more than one branch, find a sequence $T = T_0, T_1, \dots, T_k = T'$ where consecutive trees differ by a single edge swap. As we have proven, each step is canonical, and the composition of canonical transformations is canonical. Therefore, the coordinates from any two spanning trees are canonically equivalent. $\square$

Equation (8.8) is the key step. Before it, we had more variables than degrees of freedom and a non-invertible matrix in the symplectic term. After it, exactly $|T|$ independent pairs (Φ_t, Q_t) are coupled by the identity, giving the standard symplectic form $\sum_t dQ_t \wedge d\Phi_t$.

The Φ_t are the flux coordinates with the inductively homogeneous cut shifts projected out, the Q_t are the charge coordinates with the capacitive cycle circulations projected out. This is the projection done by hand, simultaneously, exactly. The spanning tree supplies the coordinates.

Remark 8.4 (Canonical pairs vs. physical degrees of freedom). The $|T|$ canonical pairs match $\text{rank}(A_C^\top)$, not $\text{rank}(M)$. By [theorem 7.5](#) and (8.4),

$$|T| = \text{rank}(M) + (|\Gamma_C| - 1), \quad (8.21)$$

so when the inductive subgraph is disconnected, $|\Gamma_C| - 1$ of the formal pairs are non-dynamical: they sit in $\ker(M) \setminus \ker(A_C^\top)$ and survive as conserved Noether charges that the Hamiltonian does not depend on. Nevertheless, they are still genuine canonical pairs and drop out automatically when H is constructed in [Section 9](#), leaving exactly $\text{rank}(M)$ dynamical pairs as expected.

9. CIRCUIT DYNAMICAL SYSTEMS ARE HAMILTONIAN

We now have everything we need. The circuit action (6.9) is Lagrangian by [theorem 6.10](#), and the spanning tree construction of [Section 8](#) has given us canonical coordinates $(\Phi_t, Q_t)_{t \in T}$. The Hamiltonian and Hamilton's equations follow almost immediately.

Before writing the Hamiltonian, we handle the constraints imposed by the null vectors of M . Because the null directions correspond to non-dynamical variables, their equations of motion impose two types of constraints:

1. For each basis vector of the homogeneous cut subspace $(\Gamma_C \cup \Gamma_L)$: the total current crossing the associated cut is conserved, which is KCL.
2. For each basis vector of the homogeneous cycle subspace $(\Delta_C \cup \Delta_L)$: the symplectic contribution vanishes, imposing a KVL-type constraint on the corresponding loop charge.

These constraints allow us to write the total energy as a function only of the canonical variables, which we call the Hamiltonian:

$$H(\Phi_t, Q_t) := E_{\text{tot}}|_{\text{constraints}}. \quad (9.1)$$

Theorem 9.1 (Circuit dynamical systems are Hamiltonian). *Let $G = (V, E, \sigma)$ be a well-posed non-dissipative planar circuit graph with smooth constitutive relations. Then the circuit dynamical system arising from G is a Hamiltonian dynamical system in the sense of [definition 2.9](#).*

Proof. We exhibit a phase space, a Hamiltonian, and a Poisson bracket satisfying [definition 2.9](#), such that the circuit equations of motion take the form $\dot{x} = \{x, H\}$.

Phase space. Fix any spanning tree $T \subseteq E_C$ of the capacitive subgraph. By [theorem 8.3](#), the circuit action (6.9) expressed in the coordinates $(\Phi_t, Q_t)_{t \in T}$ takes the form

$$S = \int dt \left[\sum_{t \in T} Q_t \dot{\Phi}_t - H(Q_t, \Phi_t) \right], \quad (9.2)$$

where H is as defined above (9.1). The phase space is $\mathcal{P} = \mathbb{R}^{|T|} \times \mathbb{R}^{|T|}$ and non-degenerate. The spanning tree is used here to exhibit explicit canonical coordinates; the same conclusion follows from M directly using only the characterization of $\ker(M)$, without choosing a tree.

Poisson bracket. By [Section A.4](#), the action (9.2) induces the canonical Poisson bracket

$$\{\Phi_t, Q_{t'}\} = \delta_{tt'}, \quad \{\Phi_t, \Phi_{t'}\} = 0, \quad \{Q_t, Q_{t'}\} = 0. \quad (9.3)$$

that is antisymmetric, non-degenerate, and bilinear. The coordinates are canonically conjugate.

Hamiltonian. The Hamiltonian is $H = E_{\text{tot}}|_{\text{constraints}}$ from (9.1). The Euler-Lagrange equations of (9.2) with respect to Φ_t and Q_t give

$$\dot{\Phi}_t = \frac{\partial H}{\partial Q_t} = \{\Phi_t, H\}, \quad \dot{Q}_t = -\frac{\partial H}{\partial \Phi_t} = \{Q_t, H\}, \quad (9.4)$$

which are exactly Hamilton's equations (2.11) that reproduce the circuit equations of motion $\Sigma_e = \Lambda_1 = \Lambda_2 = 0$ by [theorem 6.10](#). The circuit dynamical system is Hamiltonian. $\square$

9.1. Example: The Dualmon

We close by returning to the Dualmon [Figure 4](#). Both the capacitive and inductive subgraphs are connected single-edge components, so there are no homogeneous cycles ($|\Delta_C| = |\Delta_L| = 0$) and the only homogeneous cuts are the empty ones (one for each subgraph), giving $|\Gamma_L| = |\Gamma_C| = 1$. The rank of M is therefore $|V| - |\Gamma_L| - |\Gamma_C| + 1 = 2 - 1 - 1 + 1 = 1$: a single dynamical canonical pair. The unique spanning tree can be taken to be $T = \{e_{QPS}\}$ (or $T = \{e_{JJ}\}$). This produces the single canonical pair:

$$\Phi := \phi_{v_1} - \phi_{v_2}, \quad Q := q_f, \quad \{\Phi, Q\} = 1,$$

where q_f is the loop charge on the inner face (the outer face f_{ext} contributes the redundant gauge). The Hamiltonian is:

$$H = -E_Q \cos\left(\frac{2\pi Q}{2e}\right) - E_J \cos\left(\frac{2\pi \Phi}{\Phi_0}\right). \quad (9.5)$$

Hamilton's equations reproduce [\(5.7\)](#):

$$\dot{\Phi} = V_Q \sin\left(\frac{2\pi Q}{2e}\right), \quad \dot{Q} = -I_c \sin\left(\frac{2\pi \Phi}{\Phi_0}\right). \quad (9.6)$$

This is the Hamiltonian whose existence was apparent but unreachable by the methods of [Section 4.2](#). The spanning tree construction delivers it cleanly, with no ad hoc choices required.

9.2. A Reverse Reading

The story so far has run one way. We started with a circuit, translated it into a graph $G = (V, E, \sigma)$, imposed KCL, KVL, and the constitutive relations as separate axioms, and watched Euler's formula and the null-vector decomposition cooperate to deliver canonical coordinates on a spanning tree.

The same content says something rather interesting about decorated planar graphs themselves. Take any finite planar graph and color its edges by $\sigma : E \rightarrow \{C, L\}$. In other words, color each edge freely as one of two colors, with no circuit in mind at all. A priori, nothing in this setup forces the three families of constraints—two topological ([Axioms 3.10](#) and [3.12](#)), one local ([Axiom 3.13](#))—to be simultaneously satisfiable. One might imagine that there exists a clever coloring, or special planar graph substructure whose constitutive relations would over-determine the Kirchhoff system, or one that refuses to split cleanly into homogeneous cuts and loops, leaving the spanning tree construction with nothing meaningful to project onto. The content of [remark 3.20](#), [theorem 7.5](#), and [theorem 8.3](#) proves that this *never* happens for any coloring at all. Euler's formula holds for purely topological reasons, and the constitutive relations underdetermine the Kirchhoff system by exactly the two equations $\Lambda_1 = \Lambda_2 = 0$, no matter how the edges are colored.

The coloring $\sigma : E \rightarrow \{C, L\}$ is the only thing we add to the bare graph, and [theorem 7.5](#) says that this coloring alone is already enough to split node space into homogeneous cut indicators and loop space into homogeneous cycles, tied together by $|F| - |\Delta_C| - |\Delta_L| = |V| - |\Gamma_L| - |\Gamma_C| + 2$ ([corollary 7.6](#)). The spanning tree then turns that rank identity into a count of canonical pairs, $|T| = |V| - |\Gamma_L|$. The constitutive relations enter only at the very last step, attaching a smooth f_e to each edge to give it an energy.

So this paper has also presented a structural fact about decorated planar graphs: any 2-colored finite planar graph, together with any smooth choice of constitutive relations, automatically can be decomposed into homogeneous substructures, and automatically carries a Hamiltonian dynamical system in the sense of [definition 2.9](#). For some graphs this is trivial or not physically meaningful, and a purely graph-theoretic argument suffices, but nonetheless provides a different perspective.

10. CONCLUSION

Every well-posed non-dissipative circuit with smooth constitutive relations is a Hamiltonian dynamical system. Rather than construct a Hamiltonian by ansatz and verify that it reproduces the equations of motion, we have built up the structural machinery that forces the result. A circuit is translated into a graph $G = (V, E, \sigma)$, whose incidence matrix A , orientation matrix B , parity operator P , and connection matrix $M = \frac{1}{2}BPA^\top$, encode Kirchhoff's laws, homogeneous structures, the capacitor-inductor distinction, and the flux-charge symmetric pairing, respectively. The circuit action can be written in terms of the canonical coordinates of branch flux and loop charge, whose symplectic structure endows the unique Poisson bracket (Section 4.1) operation on the phase space. The Euler-Lagrange equations reproduce $\Sigma_e = \Lambda_1 = \Lambda_2 = 0$, the circuit equation of motion.

The obstruction to extracting a Hamiltonian from the Lagrangian (6.9) is precisely the null space of M . Characterizing the degeneracy has led to the identification of the right null space as gauge freedoms associated with homogeneous cuts, and the left null space with non-dynamical circulations associated with homogeneous loops. The spanning tree construction of Section 8 projects all null directions out simultaneously, and produces the canonical coordinates $(\Phi_t, Q_t)_{t \in T}$ on the reduced phase space as a result of the block-identity symplectic form. The Dualmon (Figure 4), which resists standard linear quantization approaches, falls out of the machinery with a single canonical pair and a matching periodic-in-both-variables Hamiltonian (9.5).

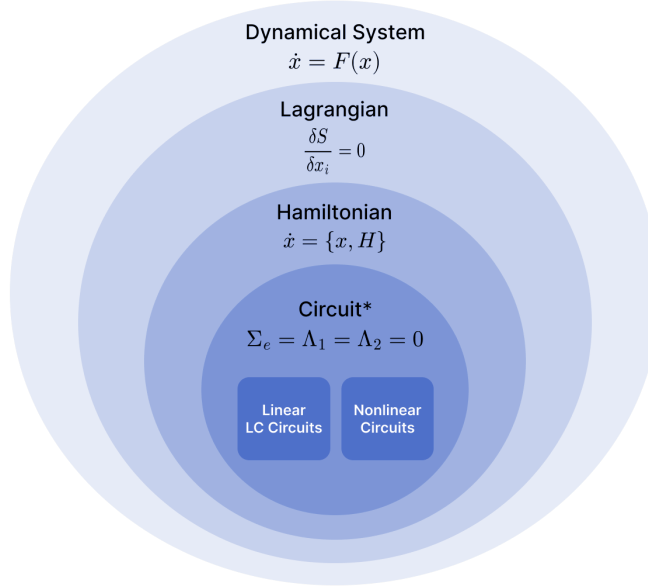


Figure 8: A domain diagram illustrating the inclusions of dynamical systems as formalized in Section 2. *A non-dissipative, well-posed, finite, planar circuit with smooth constitutive relations.

Next steps

With the classical Hamiltonian description of a superconducting circuit now in hand, canonical quantization—promoting (Φ_t, Q_t) to operators with $[\hat{\Phi}_t, \hat{Q}_{t'}] = i\hbar\delta_{tt'}$ and promoting H to $\hat{H}$ —is the subsequent procedure, and is discussed at further length in [6].

A small note on what that procedure actually involves: the single phrase “canonical quantization” hides a substantial amount of mathematical machinery (that machinery arguably being the lion’s

share of the full story). The classical construction in this paper produces a phase space and a Poisson algebra. The quantum theory is a separate discussion.

Returning to the classical result, there are many natural next directions to explore. On the side of physics, topics such as experimental circuit design can benefit from a symmetric formulation that makes duality explicit—such as engineering degrees of freedom in a large circuit. In computational quantum physics, our construction of a Hamiltonian as a local operator on a chain of conjugate pairs is the setting in which matrix product states and related tensor network methods are effective.

However, returning to mathematics, there are interesting avenues to explore in the reverse direction: what purely mathematical conclusions can be drawn about graph structures themselves by studying them through the lens of circuit symplectic geometry? Mathematics has long lent its tools to physics; the formalism proposes insights in the opposite direction.

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A. APPENDIX

A.1. Explicit Proof of Kirchhoff’s Laws in Face-Node Coordinates

Here we give a more direct proof of the sum identities (3.13) working purely from the graph-theoretic structure of A and A^* , and provide some more intuition behind this Kirchhoff reformulation.

The key observation is that $A|1\rangle = |0\rangle$: every directed edge $e = (u \rightarrow v)$ contributes -1 at its tail u and $+1$ at its head v , so each column of A sums to zero. Taking the transpose gives $\langle 1|A^\top = \langle 0|$, i.e. left-multiplication of $A^\top$ by the row of ones annihilates it.

1. The identity $\sum_v \dot{\phi}_v = 0$. By KVL (Axiom 3.10), the branch voltages are $\dot{\phi}_e = (A^\top |\dot{\phi}_V\rangle)_e$. Summing over all nodes on the left-hand side of the branch-flux assignment and using $\langle 1|A^\top = \langle 0|$:

$$\sum_{v \in V} \dot{\phi}_v = \langle 1 | \dot{\phi}_V \rangle, \quad \langle 1 | A^\top | \dot{\phi}_V \rangle = 0.$$

The uniform direction $|1\rangle \in \ker(A^\top)$ therefore corresponds to a pure gauge freedom in node-flux space (shifts $|\phi_V\rangle \mapsto |\phi_V\rangle + c|1\rangle$ leave all branch fluxes unchanged), and we may fix this gauge by demanding $\langle 1 | \dot{\phi}_V \rangle = 0$, yielding $\sum_v \dot{\phi}_v = 0$.

2. The identity $\sum_f \dot{q}_f = 0$. The dual statement proceeds analogously. Since $A^*|1\rangle = |0\rangle$ on the dual graph (each edge bounds two faces with opposite orientation), we have $\langle 1 | A^{*\top} = \langle 0|$. By KCL expressed in face charges, the branch currents are $I_e = \dot{q}_e = (A^{*\top} |\dot{q}_F\rangle)_e$, so

$$\langle 1 | A^{*\top} | \dot{q}_F \rangle = 0,$$

and the uniform shift $|q_F\rangle \mapsto |q_F\rangle + c|1\rangle$ on face charges is a gauge freedom which we fix by setting $\sum_f \dot{q}_f = 0$.

Both identities are thus direct consequences of the fact that the column sums of A (and of A^*)

vanish, together with the gauge-fixing choice. At a quick glance, this reformulation looks almost tautological, but it takes a moment to see what is really being said. Naively, KVL is one equation per loop and KCL is one per node, yet here they collapse to just two scalar conditions. Claiming that currents in each individual loop must sum to zero is distinct from saying the loop charges from every face sum to zero. The former is an external constraint on physical currents, while the latter is an identity that emerges once currents are expressed in terms of potentials. So, the key point is that the constraints have migrated into the coordinates themselves. Once we write $\phi_e = (A^\top |\phi_V\rangle)_e$ and $I_e = (A^{*\top} |\dot{q}_F\rangle)_e$, KVL and KCL are satisfied automatically by the exactness relations, because a signed sum of node-differences around any loop telescopes to zero.

A.2. Symplectic Structure and the Origin of the Poisson Bracket

The purpose of this appendix is to derive the Poisson bracket systematically from the symplectic structure of the Lagrangian. The purpose is to motivate why the Poisson bracket is a natural (and in fact unique) choice. The proof is outlined below:

Proof. Let $x = (x_1, \dots, x_{2n})$ be phase-space coordinates on a $2n$ -dimensional space, and suppose the equations of motion arise from an action

$$S = \int d\tau (\omega_{\alpha\beta} x_\alpha \dot{x}_\beta - E(x)), \quad (\text{A.1})$$

where $\omega_{\alpha\beta}$ is a real matrix of constants and $E(x)$ is the energy. Since the symmetric part of ω contributes only a total time derivative, we take ω antisymmetric without loss of generality.

Deriving the equations of motion. Varying (A.1):

$$\delta S = \int d\tau (\omega_{\alpha\beta} \delta x_\alpha \dot{x}_\beta + \omega_{\alpha\beta} x_\alpha \delta \dot{x}_\beta - \frac{\partial E}{\partial x_\alpha} \delta x_\alpha).$$

Integrating the middle term by parts and using antisymmetry of ω :

$$\omega_{\alpha\beta} \int x_\alpha \delta \dot{x}_\beta d\tau = \omega_{\alpha\beta} \int \dot{x}_\beta \delta x_\alpha d\tau.$$

Collecting terms, stationarity $\delta S = 0$ for all variations gives

$$2\omega_{\alpha\beta} \dot{x}_\beta = \frac{\partial E}{\partial x_\alpha}. \quad (\text{A.2})$$

Inverting to find the bracket. When ω is invertible,

$$\dot{x}_\beta = \frac{1}{2} (\omega^{-1})_{\beta\alpha} \frac{\partial E}{\partial x_\alpha}. \quad (\text{A.3})$$

Finally, defining the following completes the proof:

$$\{f, g\}_\omega := (\omega^{-1})_{\alpha\beta} \frac{\partial f}{\partial x_\alpha} \frac{\partial g}{\partial x_\beta} \quad (\text{A.4})$$

Notice that (A.3) becomes $\dot{x} = \{x, H\}_\omega$ where $H = \frac{1}{2}E$. The bracket is antisymmetric because the inverse of an antisymmetric invertible matrix is antisymmetric.

Eventually after the spanning tree reduction of Section 8, the circuit action takes the form $S = \int d\tau [\sum_t Q_t \dot{\Phi}_t - H]$. The symplectic matrix is thus (after reading off the canonical 1-form

and taking the exterior derivative)

$$\omega = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}, \quad \omega^{-1} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix},$$

giving exactly the bracket $\{f, g\} = \langle \nabla_{\Phi} f | \nabla_Q g \rangle - \langle \nabla_Q f | \nabla_{\Phi} g \rangle$ from [definition 4.1](#). $\square$

A.3. Spanning Trees on the Capacitive Subgraph

We collect here the elementary facts about the spanning trees (sometimes *forests*) of (V, E_C) used in [Section 8](#): that one always exists, that it has exactly $|V| - |\Gamma_L|$ branches, and that the corresponding co-tree has $|\Delta_C|$ branches. The arguments are rather standard graph theory facts; we produce them here to make the connection to the homogeneous-cut and homogeneous-loop counts of [theorem 7.5](#) explicit.

Recall from [theorem 7.5](#) and the surrounding discussion that $|\Gamma_L|$ is the dimension of $\ker(A_C^\top)$, equivalently the number of connected components of the capacitive subgraph (V, E_C) . Each such component corresponds to one inductively homogeneous cut, namely the cut whose indicator is constant on the component and zero elsewhere. In the language of [Section A.5](#), these components are the inductively shunted islands.

Branch count. Let $\Gamma_L = \{i_1, \dots, i_{|\Gamma_L|}\}$ enumerate the connected components of (V, E_C) , and write V_k for the vertex set of component i_k , so that $V = \cup_k V_k$. A spanning tree restricted to component i_k is a connected acyclic subgraph on $|V_k|$ vertices, and any such subgraph has exactly $|V_k| - 1$ edges (by induction on $|V_k|$, or from Euler's formula on a tree). Summing over components,

$$|T| = \sum_{k=1}^{|\Gamma_L|} (|V_k| - 1) = |V| - |\Gamma_L|. \quad (\text{A.5})$$

Therefore, when (V, E_C) is connected, $|\Gamma_L| = 1$ and $|T| = |V| - 1$. When (V, E_C) has isolated nodes (vertices touched only by inductors, or not touched at all), each contributes a component with $|V_k| = 1$ and zero edges, these are essentially ignored by the spanning tree.

Co-tree count. The co-tree $E_C \setminus T$ contains

$$|E_C \setminus T| = |E_C| - |T| = |E_C| - |V| + |\Gamma_L|. \quad (\text{A.6})$$

This count equals $|\Delta_C|$, the number of independent capacitively homogeneous cycles. Each co-tree branch $e \in E_C \setminus T$ closes a unique fundamental cycle with T , and this cycle is supported entirely on E_C by construction. So, the fundamental cycles are linearly independent and span the cycle space of (V, E_C) , which has dimension $|\Delta_C|$.

Existence by greedy construction. A spanning tree of (V, E_C) always exists. Construct one as follows:

1. Start with $T = \emptyset$ and the vertex set V .
2. Examine the capacitive branches E_C in any order. For each branch $e \in E_C$, add e to T if and only if doing so does not create a cycle.
3. Terminate when all of E_C has been examined.

The proof of which is left to the reader to convince themselves of, as this is a common argument in graph theory. Note that the construction also exposes the non-uniqueness of T . This is the freedom that [theorem 8.3](#) (Part 3) shows is irrelevant.

A.4. Canonical Conjugate Pairs from the Spanning Tree Action

After the spanning tree reformulation of [theorem 8.3](#), the action is

$$S = \int d\tau \left[\sum_{t \in T} Q_t \dot{\Phi}_t - H(\Phi_t, Q_t) \right].$$

As reasoned in [Section A.2](#), the symplectic matrix is ω_{IJ} with $I = (t, \Phi)$, $J = (t', Q)$:

$$[\omega_{IJ}] = \begin{pmatrix} 0 & -I_{|T|} \\ I_{|T|} & 0 \end{pmatrix}. \quad (\text{A.7})$$

This is antisymmetric and non-degenerate. The associated symplectic 2-form is $\omega_{2\text{-form}} = \sum_{t \in T} dQ_t \wedge d\Phi_t$. By [\(A.4\)](#) the Poisson bracket under indices $I = (t, \Phi)$, $J = (t', Q)$ is (for the sake of explicit expansion):

$$\{F, G\} = \sum_{I, J} (\omega^{-1})^{IJ} \partial_I F \partial_J G \quad (\text{A.8})$$

$$= \sum_{t, t' \in T} (\omega^{-1})^{\Phi_t \Phi_{t'}} \partial_{\Phi_t} F \partial_{\Phi_{t'}} G + \sum_{t, t' \in T} (\omega^{-1})^{\Phi_t Q_{t'}} \partial_{\Phi_t} F \partial_{Q_{t'}} G \quad (\text{A.9})$$

$$+ \sum_{t, t' \in T} (\omega^{-1})^{Q_t \Phi_{t'}} \partial_{Q_t} F \partial_{\Phi_{t'}} G + \sum_{t, t' \in T} (\omega^{-1})^{Q_t Q_{t'}} \partial_{Q_t} F \partial_{Q_{t'}} G \quad (\text{A.10})$$

$$= \sum_{t, t' \in T} \delta_{tt'} \partial_{\Phi_t} F \partial_{Q_{t'}} G - \sum_{t, t' \in T} \delta_{tt'} \partial_{Q_t} F \partial_{\Phi_{t'}} G \quad (\text{A.11})$$

$$= \sum_t [(\partial_{\Phi_t} F)(\partial_{Q_t} G) - (\partial_{Q_t} F)(\partial_{\Phi_t} G)]. \quad (\text{A.12})$$

Reading off the special cases:

$$\begin{aligned} \{\Phi_t, Q_{t'}\} &= \delta_{tt'}, \\ \{\Phi_t, \Phi_{t'}\} &= 0, \\ \{Q_t, Q_{t'}\} &= 0. \end{aligned}$$

The symplectic matrix [\(A.7\)](#) is actually already in the canonical block form of [remark 2.10](#). Introducing a bit of terminology here, this is the statement that $(\Phi_t, Q_t)_{t \in T}$ are *Darboux coordinates* and the explicit expansion can be skipped to directly conclude conjugation. Regardless, the canonical relations follow directly from the spanning tree structure of the action.

A.5. The Ω -Based Null Space Characterization (For Reference)

For completeness, and to connect with the formulation of [\[6\]](#), we record the null space theorem for the reduced capacitive incidence matrix

$$\Omega := A_C^\top \in \mathbb{R}^{|E_C| \times |V|}. \quad (\text{A.13})$$

The theorem itself is correct and both formulations (the Ω -based and the M -based) quantize the same class of circuits. What differs is structural—the Ω -approach lacks the flux-charge symmetry of M , and non-dynamical directions that M absorbs as null vectors show up in the Ω -formulation as separate Noether currents that must be handled by hand. [Figure 5](#) illustrates how asymmetrically depending on capacitive edges can hide circuit structure.

Definition A.1 (Inductively Shunted Island). An inductively shunted island is a connected component

of the undirected graph (V, E_C) —the capacitive subgraph. We write $\Gamma_L^{(\Omega)}$ for the set of all such islands, with indicator ket $|1_i\rangle$ for island i .

Definition A.2 (Capacitive Loop). A capacitive loop is a closed walk in (V, E_C) . We write $\Delta_C^{(\Omega)}$ for all such loops, with signed indicator ket $|s_\delta\rangle$ for loop δ .

Theorem A.3 (Null vectors of Ω). *Let Ω be the reduced capacitive incidence matrix (A.13).*

1. $\ker(\Omega) = \text{span}\{|1_i\rangle : i \in \Gamma_L^{(\Omega)}\}$: vectors constant on each inductively shunted island.
2. $\ker(\Omega^\top) = \text{span}\{|s_\delta\rangle : \delta \in \Delta_C^{(\Omega)}\}$: signed sums of capacitive loop indicators.
3. $\text{rank}(\Omega) = |V| - |\Gamma_L^{(\Omega)}|$.

Proof. We prove each part, working in the same bra-ket conventions as [theorem 7.5](#) and freely using the exactness relation (3.9), the Exactness Lemma [lemma 7.1](#), and the column-sum identity $A|1\rangle = |0\rangle$ from [lemma 3.14](#). The overall structure mirrors the proof of [theorem 7.5](#), with the simplification that Ω sees only capacitive edges so no edge-parity operator is needed.

Part 1 (right null vectors).

($\Leftarrow$) Suppose $|\phi_V\rangle = \sum_i \beta_i |1_i\rangle$. For any capacitive edge $e = (u \rightarrow v) \in E_C$, both endpoints u and v lie in the same inductively shunted island (by definition of an island as a connected component of (V, E_C)), and therefore receive the same coefficient β_i . Thus

$$(\Omega|\phi_V\rangle)_e = \phi_v - \phi_u = \beta_i - \beta_i = 0$$

for every $e \in E_C$, giving $\Omega|\phi_V\rangle = |0\rangle$.

($\Rightarrow$) Suppose $\Omega|\phi_V\rangle = |0\rangle$, equivalently $A_C^\top|\phi_V\rangle = |0\rangle$. For each capacitive edge $e = (u \rightarrow v)$,

$$(A_C^\top|\phi_V\rangle)_e = \phi_v - \phi_u = 0,$$

so $\phi_u = \phi_v$. Iterating along paths in the capacitive subgraph (V, E_C) forces $|\phi_V\rangle$ to be constant on every connected component of (V, E_C) . Writing $|\phi_V\rangle$ in the basis of island indicators yields the representation $|\phi_V\rangle = \sum_i \beta_i |1_i\rangle$.

Part 2 (left null vectors).

($\Leftarrow$) Let $|\ell\rangle = |s_\delta\rangle$ for a single capacitive loop δ . Write $\delta = (e_1, \dots, e_k)$ as an ordered closed walk with orientation signs $\sigma_1, \dots, \sigma_k \in \{\pm 1\}$ relative to the graph orientation, so that $(s_\delta)_{e_i} = \sigma_i$. Let $v_1, \dots, v_k$ be the nodes visited, with $v_{k+1} = v_1$. Then

$$\langle s_\delta | \Omega = \sum_{i=1}^k \sigma_i (\langle v_{i+1} | - \langle v_i |), \quad (\text{A.14})$$

where the i -th term picks up $+\langle v_{i+1} |$ at the head and $-\langle v_i |$ at the tail of e_i (reversed if $\sigma_i = -1$, but the traversal convention absorbs the sign consistently). The sum telescopes around the closed loop to zero. Any linear combination $|\ell\rangle = \sum_\delta \alpha_\delta |s_\delta\rangle$ therefore also satisfies $\langle \ell | \Omega = \langle 0 |$.

($\Rightarrow$) Suppose $\langle \ell | \Omega = \langle 0 |$. The condition reads, for every node $v \in V$,

$$\sum_{e \in E_C} \ell_e \Omega_{e,v} = 0.$$

Recall $\Omega_{e,v} = (A_C^\top)_{e,v}$ is +1 if e starts at v , -1 if e ends at v , and 0 otherwise. So this says the signed sum of ℓ_e over all capacitive edges incident to v vanishes. Interpreting $|\ell\rangle \in \mathbb{R}^{|E_C|}$ as a weighted capacitive chain $\sum_e \ell_e [e]$, the condition is that its boundary vanishes at every vertex—i.e., $|\ell\rangle$ is a cycle of the capacitive subgraph.

Formally, we apply the Exactness Lemma [lemma 7.1](#) specialized to the capacitive subgraph. Take $X = \mathbb{R}^{|V|}$, $Y = \mathbb{R}^{|E_C|}$, $Z = \mathbb{R}^{|V|}$, with maps $A \mapsto A_C^\top : X \rightarrow Y$ and $B \mapsto A_C : Y \rightarrow Z$, and P the identity (no edge-parity is needed since Ω only sees one edge type). The exactness hypothesis $\ker(A_C) = \text{Im}(A_C^\top)$ holds on the capacitive subgraph by the usual graph-theoretic exactness [\(3.9\)](#). Then $\langle \ell | \Omega = \langle 0 |$ is equivalent to $A_C |\ell\rangle = |0\rangle$, so $|\ell\rangle \in \ker(A_C)$, which is the cycle space of (V, E_C) . Expanding in a basis of independent capacitive cycles gives $|\ell\rangle = \sum_{\delta \in \Delta_C^{(\Omega)}} \alpha_\delta |s_\delta\rangle$.

Part 3 (rank).

Part 1 identified $\ker(\Omega) = \text{span}\{|1_i\rangle : i \in \Gamma_L^{(\Omega)}\}$, and the island indicators $\{|1_i\rangle\}$ are linearly independent (they have disjoint supports). Therefore $\dim \ker(\Omega) = |\Gamma_L^{(\Omega)}|$, and by rank nullity,

$$\text{rank}(\Omega) = |V| - |\Gamma_L^{(\Omega)}|. \quad (\text{A.15})$$

□

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